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Loan guarantees, bank lending and credit risk reallocation ☆

Carlo Altavilla [a], Andrew Ellul [b,*], Marco Pagano [c], Andrea Polo [d], Thomas Vlassopoulos [e]

[a] European Central Bank, CSEF and CEPR, Germany
[b] Indiana University, CSEF, CEPR and ECGI, United States of America
[c] University of Naples Federico II, CSEF, EIEF, CEPR and ECGI, Italy
[d] Luiss University, EIEF, CEPR and ECGI, Italy
[e] European Central Bank, Germany

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ABSTRACT

Do banks extending government-guaranteed loans simultaneously reduce their risk exposure to firms? Using unique euro-area credit register data and the COVID-19 guarantee programs as a laboratory, we find that 1 euro of guaranteed lending was associated with a reduction of 28 cents in non-guaranteed credit, relative to other banks lending to the same firm. Substitution was highest for riskier and smaller firms in more affected sectors and for stronger banks. Nevertheless, banks offered cheaper credit and longer maturities to guaranteed loan recipients, especially more fragile ones. This improvement in lending terms is the flipside of credit substitution.

1. Introduction

Loan guarantees are well-established policy tools: governments rely on them to encourage bank lending by shouldering borrowers' default risk. Their typical rationale is to overcome frictions leading to the under-provision of credit to particular types of firms, especially during economic crises when default waves may propagate across debt chains interconnecting firms (Glode and Opp, 2023), leading even otherwise viable firms to be liquidated (Antill and Clayton, 2025). Loan guarantees provide the required backstop even to hard-hit firms, insofar as transferring default risk to the government encourages banks to increase lending. They may also be a faster and more efficient way to support firms than direct government funding, as typically banks' screening technologies and established relationships endow them with better information than the government about the quality of each firm: by leveraging banks' knowledge provided they retain skin in the game, liquidity should more likely reach viable firms than if the government were to pick the firms to be saved and those to be liquidated (Philippon, 2021).

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* Corresponding author.
E-mail addresses: carlo.altavilla@ecb.int (C. Altavilla), anellul@indiana.edu (A. Ellul), pagano56@gmail.com (M. Pagano), andreapolo@gmail.com (A. Polo), thomas.vlassopoulos@ecb.int (T. Vlassopoulos).

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Thus, the efficiency of such programs rests on bank lending being highly responsive to public guarantees. This is reduced if the banks providing guaranteed loans simultaneously reduce their non-guaranteed loans or credit lines to the same debtors, so as to decrease their credit risk exposure toward them by shifting it to the government. The main objective of this paper is to assess the extent to which banks engage in such "credit substitution", thus reducing public guarantees' effectiveness in expanding credit. For this purpose, we exploit the COVID-19 guarantee programs as our laboratory. Bank loan guarantees were massively used to stabilize the economy in response to the pandemic shock and the resulting dry-up in firms' liquidity. The possibility of credit substitution was indeed on policymakers' minds when laying out eligibility guidelines for these programs, as we explain in Section 3,[1] and the media was also aware of such risk.[2] In this respect, the pandemic can be regarded as a testing ground to study the effectiveness of a massive expansion of bank loan guarantees amid a major recession. In this study we consider how loan guarantees affected the overall conditions at which banks offered credit: not only the amount of lending, but also the interest rates and the maturities at which it was made available, taking into account the trade-offs between these dimensions of credit contracts.

At the aggregate level, net lending in the euro area grew less than one-for-one with the expansion of guaranteed loans between April and August 2020, i.e. during the first wave of the pandemic and following the launch of loan guarantee programs, as shown by Fig. 1. Of course, such a macro-level correlation is *per se* no evidence that credit substitution occurred upon the introduction of loan guarantee schemes. Hence, this paper bases the analysis on firm-bank matched data drawn from the euro-area Anacredit credit register.

The granularity and breadth of our data enables us to investigate various facets of the issue of substitution. First, data for firms with multiple lending relationships help us to address the issue of firms' selection into loan guarantee programs. Second, the data enable us to investigate how credit substitution varied across firms and banks, depending on their characteristics, and whether it differed across countries adopting different loan guarantee programs. Finally, we can investigate whether the provision of public guarantees was associated with changes in the cost of credit and/or loan maturities and whether these were related to credit substitution.

To guide the empirical analysis, we propose a model of the effects of public loan guarantees on banks' non-guaranteed lending and interest rates. The model shows that loan guarantees should be expected to have different effects across firms depending on their credit risk and bargaining power in renegotiating credit with their initial lender.

The model predicts that, by opening an avenue to transfer credit risk to the government, loan guarantees induce banks to reduce non-guaranteed credit more to riskier firms than to safer ones. Moreover, if banks are competitive, credit substitution is predicted to correlate with a drop in the average interest rate charged to the relevant borrowers, who will thereby appropriate the benefit of the public guarantee. This benefit is greater for high-risk clients, who enjoy a sharper credit cost reduction. This generates a positive correlation across firms between credit substitution and interest rate reduction upon the introduction of loan guarantees. However, the drop in credit costs will be lower if frictions in renegotiating loans enable the initial lender to share in the benefit from the guarantee via greater profits.

To take these predictions to the data, we measure credit substitution as (the negative of) the change in non-guaranteed credit in the pandemic period relative to its pre-pandemic level.[3] To estimate the relationship between credit substitution and the supply of guaranteed loans, one must consider that the firms that received guaranteed loans are not a random sample, but one whose composition depends on eligibility rules. To overcome this selection issue, we exploit variation across firm-bank relationships for firms that received guaranteed loans, rather than variation across firms in the whole sample: for firms that borrow from multiple banks, we compare the change in non-guaranteed exposures between banks extending guaranteed loans and other banks lending to the *same* firm, using a methodology akin to that of Khwaja and Mian (2008). Exploiting such within-firm variation enables us to address the counterfactual problem of whether firms that experienced substitution upon receiving guaranteed loans would have faced a credit cut anyway. This concern is especially relevant for firms that were already financially weaker before the pandemic (but still in line with eligibility rules) and/or were severely hit by the shock: had they not received guaranteed loans, they would have been likely to face a credit reduction anyway. Our approach addresses this concern by effectively considering substitution by banks that did not extend guaranteed loans as the counterfactual credit response for banks that extended guaranteed loans. Exploiting such within-firm variation, we find that 1 euro of guaranteed lending is associated with a 28-cent reduction in non-guaranteed credit by the bank offering the guaranteed loan relative to other banks lending to the same firm.

Estimates also show that credit substitution was largest for guaranteed lending to riskier and smaller firms in sectors and countries more affected by the pandemic, and borrowing from stronger banks, i.e., those with less NPLs. These results align with the model's predictions, insofar as these firms have greater default probability and these banks have higher bargaining power vis-à-vis the corresponding borrowers. The regressions also include bank-level fixed effects to control for unobserved bank heterogeneity in the provision of credit during the pandemic.[4] These results are qualitatively consistent across the four largest euro-area countries, although the extent of substitution differs across them, being largest in Italy and lowest in Germany.

A potential concern with our estimates is that observed credit substitution could mechanically reflect the expiration of a previous loan and its replacement with the guaranteed loan, rather than a deliberate decision to reduce credit. However, several tests show that substitution cannot be ascribed to this mechanical effect from the expiration of pre-existing non-guaranteed credit. Moreover, we find that credit substitution is significantly stronger for credit issued via revolving credit lines, which can be revoked unilaterally by the bank at any time, than for term loans and non-revolving credit lines, whose terms can be altered only by renegotiating them with clients.

Our results on firms with multiple bank relationships may be subject to two selection issues. First, banks issuing guaranteed loans are not randomly assigned among the lenders of a given firm. Second, the characteristics of firms with multiple bank relationships may differ

[1] Blanchard et al. (2020) describe this possible problem as follows: "The main danger is the transfer of pre-existing exposures. A bank with an exposure to a firm could ask it to use the guaranteed debt to repay its existing loans. This would be a transfer of risk to the state".

[2] For instance, the Financial Times drew attention to Greensill Bank AG using state-backed loans from three European governments to reduce its exposure to distressed companies owned by metal magnate Sanjeev Gupta (see "Greensill used taxpayer loans to cut exposure to Sanjeev Gupta", Financial Times, 4 July 2021); similarly, Italian and Spanish newspapers flagged the risk that loan guarantees may end up shielding banks more than firms hit by the pandemic shock (see "Lo scudo delle garanzie fiscali copre più le banche delle imprese", La Repubblica, 1 March 2021 and "Una parte de los créditos avalados por el ICO para rescatar a las pymes se queda en manos de la banca para cubrir deudas de los empresarios", El Diario, 13 May 2020).

[3] Given this definition of credit substitution, we exclude from the analysis firms that access the credit market for the first time in the wake of the adoption of the loan guarantee program.

[4] For example, some banks may have lent more aggressively during the guarantee program because they had better information technology capabilities to handle the high number of applications (Core and De Marco, 2023).

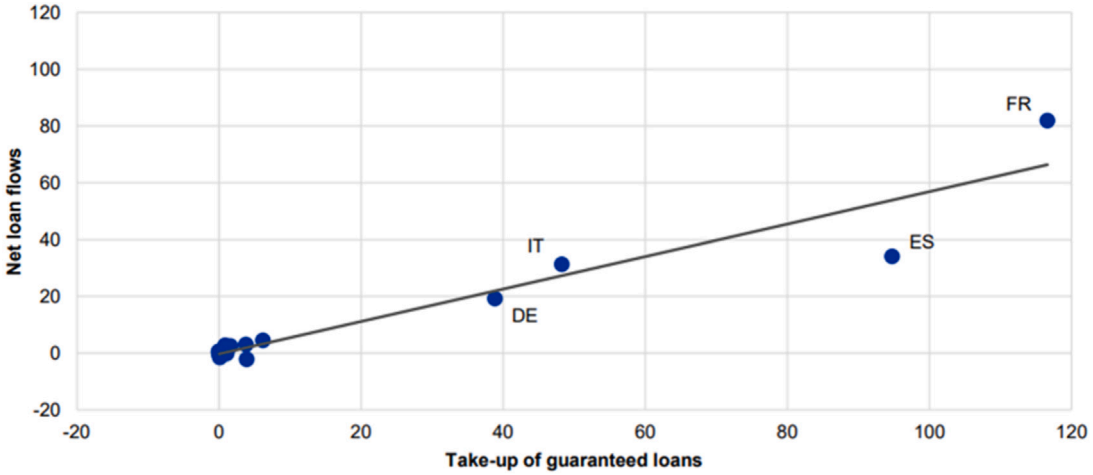


Fig. 1. Guaranteed loans and net lending: country-level data (billion euros)
This figure shows the cross-country relation between the amount of take-up of guaranteed loans and the net loan flows at country level, over the period April-August 2020. Each dot refers to a country in the euro area. Data sources: Kreditanstalt für Wiederaufbau for Germany, Instituto de Crédito Oficial for Spain, Ministère de l'Économie et des Finances for France, Ministero dell'Economia e delle Finanze and Banca d'Italia for Italy, various national authorities for other euro area countries, news sources, ECB and ECB calculations.

from those of the population. To sign the first potential selection bias, we investigate the characteristics of banks issuing guaranteed loans relative to other banks lending to the same firm, and find that banks with more capital, greater size and tighter relationships with the firm were more likely to extend guaranteed credit.[5] As these banks' characteristics are typically associated with a greater supply of credit at times of stress (Bolton et al., 2016, and Jiménez et al., 2012), our results, if anything, underestimate the extent of substitution by banks issuing guaranteed credit. To face the second possible selection issue, we expand the sample to include single-bank firms and estimate regressions that do not include firm fixed effects (as in the main specification) but include industry-location-size fixed effects. The estimates obtained using this expanded sample are qualitatively similar to our baseline ones, except that the estimates of credit substitution are approximately 2 to 6% higher.

As our model predicts that loan guarantees should affect not only the availability of credit, but also other lending conditions offered to the firm, we then investigate how the provision of guarantees correlates with changes in the interest rates and loan maturities offered by banks providing guaranteed loans, considering the conditions that apply to their overall lending to the firm, i.e., both guaranteed and non-guaranteed credit. We find that the bank providing guaranteed credit reduces the average interest rate and lengthens the average maturity of its credit relative to other banks lending to the same firm without guarantee, and these effects are stronger for weaker firms, i.e., those in more affected sectors and countries, with smaller size and/or a larger fraction of arrears. These results align with the predictions of our model, which predicts that riskier firms obtain a larger reduction in the cost of credit upon receiving guaranteed loans. Hence, in line with the model, the banks granting guaranteed loans offer more generous interest rate concessions to their riskier clients while reducing their non-guaranteed exposure to them by more. As a result, at the cross-sectional level, the availability of loan guarantees generates a tradeoff: in bank-firm relationships where firms are faced with more credit substitution, they obtain comparatively more generous interest rate concessions. Consistent with the model, we find that this tradeoff is affected by the distribution of bargaining power between banks and firms: when the bank providing the guaranteed loan is the sole firm's lender rather than one of several lenders, it grants a much smaller interest rate concession to the firm for a given amount of credit substitution. Similarly, when the bank lent to the firm via a callable credit line, it grants a smaller interest rate concession upon extending a guaranteed loan than when it provided no callable credit to the firm, so that substitution requires renegotiating its non-guaranteed credit.

Finally, we ask whether the substitution observed at bank-firm level is also present at firm level. As these regressions rely on firm-level variation rather than on within-firm variation, they also include single-bank firms. In firm-level estimates, larger guaranteed loans are associated with a smaller degree of substitution, 1 euro of additional guarantees being associated, on average, with a drop of 9 cents in non-guaranteed lending. This indicates that the drop in the supply of non-guaranteed credit by banks that provide the guaranteed loan is mitigated by the behavior of the banks that do not issue the guaranteed loan. Also at firm level, substitution is confirmed to be stronger for weaker firms and for stronger banks.

The overall thrust of our results is that in the euro area, government guarantees contributed to the continued extension of credit to relatively creditworthy firms hit by the pandemic. However, they also triggered a certain amount of credit substitution, by which pre-existing credit risk was merely transferred from banks to taxpayers. This was not necessarily a policy failure, for three reasons. First, absent such programs, banks could have reduced their exposures even more, generating default waves that might have crippled otherwise viable firms. Second, by using these programs to de-risk their balance sheets, banks may have preserved more lending capacity to face the post-pandemic recovery, reducing the risk of a cliff-effect credit crunch upon program termination. Third, since substitution moderated lending to the riskiest firms, these should have accumulated less leverage than in a counterfactual world where no substitution occurred (Brunnermeier and Krishnamurthy, 2020).

Our paper contributes to three recent strands of research. The first is the literature on the effectiveness of public guarantee programs. Bachas et al. (2021) investigate the effectiveness of guarantees provided by the Small Business Agency in the U.S. and find significant bunching in the loan size distribution at thresholds where guarantee generosity decreases, showing that lenders prefer to issue loans with higher guarantee rates. After our work, other papers have focused on evidence from individual countries and particular aspects of Europe's

[5] This confirms the importance of healthy balance sheets and relationships as key mechanisms in liquidity provision at times of stress, consistently with U.S. evidence by Li and Strahan (2021).

pandemic loan guarantee programs.[6] Our contribution relative to this literature is to provide the first systematic investigation of the effects of these programs on the entire gamut of credit conditions in bank-firm relationships, exploring how these effects vary depending on firm and bank characteristics and taking into account the relationship between credit substitution and cost of credit.[7] A further contribution is to show that our results hold across countries with different program designs and institutional characteristics.

Our paper also contributes to the literature on the variation of bank liquidity provision across firm sizes at times of stress. Previous work documents that, at the inception of the pandemic, large firms raised liquidity by drawing down their credit lines (Acharya and Steffen, 2020; Li et al., 2020); instead, small and medium enterprises (SMEs), which were prevented from doing so, relied on guaranteed loans to mitigate their liquidity shortfall (Greenwald et al., 2020; Kapan and Minoiu, 2021; Chodorow-Reich et al., 2022). Our paper adds to this literature by investigating how the effects of loan guarantee programs varied depending on firm characteristics, as smaller and riskier firms faced more severe credit substitution but benefited from greater interest rate reductions and maturity extensions. This underscores the importance of providing a complete picture of the effects of such programs by investigating all lending conditions rather than solely the amount of lending.

Finally, our paper is relevant for the effectiveness of policies aimed at counteracting the real effects of the pandemic. Our evidence for Europe differs from that for the United States, where Granja et al. (2022) find that the funds of the Paycheck Protection Program (PPP) were not channeled to the worst-hit sectors, and Cororaton and Rosen (2021) document that PPP targeted mostly firms with higher leverage, less cash and worse business prospects.[8]

The paper is organized as follows. Section 2 proposes a model to guide the empirical analysis. Section 3 describes the institutional framework of euro-area loan guarantees. Sections 4 and 5 present the data and the empirical methodology. Section 6 reports and discusses the results. Section 7 concludes.

2. A model of loan guarantees and credit substitution

This model is intended to guide the empirical analysis regarding the effects of the introduction of loan guarantees on the amount and cost of non-guaranteed credit. It shows how these effects vary depending on firms' default risk and their bargaining power in renegotiating pre-existing loans.

First, upon extending guaranteed loans, banks are predicted to reduce non-guaranteed loans to riskier borrowers more than to comparatively safe ones; indeed, the latter may even obtain more non-guaranteed lending, as the guarantee induces banks to relax their borrowing constraints. Hence, credit substitution should correlate positively with firm risk.

Second, the model predicts that if non-guaranteed loans can be renegotiated competitively, guaranteed loans trigger a drop in firms' average cost of credit, especially for riskier borrowers. Hence, the drop in the cost of credit is predicted to correlate positively with credit substitution across firms. This interest rate response is muted if firms have low bargaining power in renegotiating pre-existing loans so that initial lenders appropriate most of the guarantee's expected benefit.

2.1. Setup

There is a continuum of firms, each of which can undertake one of two investment projects that differ in size and payoff: small projects have an initial cost I_s and pay Y if they succeed and zero otherwise, while large projects cost an amount $I_\ell > I_s$ and pay X if successful and zero otherwise. Firms are assumed to have no initial assets and must fund their projects entirely out of bank credit. Investors feature risk neutrality and no discounting.

Firms differ in their failure probability: projects run by good (g) and bad (b) firms fail with probabilities p and $p + \Delta$, respectively. Small projects have positive NPV even if carried out by a bad firm, i.e., $(1 - p - \Delta)Y > I_s$, while large projects have negative NPV even if undertaken by a good firm, i.e., $(1 - p)X < I_\ell$. However, large projects generate a non-pledgeable payoff B, due to moral hazard arising from limited enforcement or private benefit extraction by firm owners. This non-pledgeable payoff is large enough that good firms would want to undertake a large project rather than a small one if they could fund it internally:

$$B + (1 - p)(X - Y) > I_\ell - I_s. \tag{1}$$

We analyze two settings: one with no public guarantees and another where guarantees become unexpectedly available after financing is issued. The guaranteed loan, L^G, is large enough to make funding the large project viable, but only if managed by a good firm:

$$(1 - p)X > I_\ell - L^G > (1 - p - \Delta)X, \tag{2}$$

where $I_\ell - L^G$ is the part of the project's cost not covered by the guaranteed loan. The rationale for this assumption is that the loan guarantee program is designed to avoid shifting excessive credit risk onto the government: the guarantee is sufficient to relax banks' credit standards vis-à-vis good risks but not bad ones. For the same reason, the guaranteed loan is assumed not to exceed the cost of the small project, i.e., $L^G \leq I_s$.

2.2. Equilibrium with no guaranteed loans

Absent loan guarantees, only small projects are funded in equilibrium, as large ones have negative NPV. If the credit market is perfectly competitive, good and bad firms obtain term loans of amount I_s with pledged repayments F_{sg} and F_{sb}, respectively:

$$F_{sg} = \frac{I_s}{1 - p}, \quad F_{sb} = \frac{I_s}{1 - p - \Delta}, \tag{3}$$

implying pledged interest rates $\frac{p}{1-p}$ and $\frac{p+\Delta}{1-p-\Delta}$. The value of debt issued to the two firm types equals the small project's cost: $D_{sg} = D_{sb} = I_s$, and the expected firm payoffs equal the NPVs of their respective projects:

$$\Pi_{sg} = (1-p)(Y-F_{sg}) = (1-p)Y - I_s, \quad \Pi_{sb} = (1-p-\Delta)(Y-F_{sb}) = (1-p-\Delta)Y - I_s. \tag{4}$$

[6] Cascarino et al. (2022) study their effectiveness in Italy by exploiting variation in the fractional guarantees offered by different programs and find that programs with higher fractional guarantees triggered a greater expansion in credit. Jiménez et al. (2022) analyze the effect of relationship lending on guaranteed loans in Spain: they show that pre-existing relationships affect both the allocation of guaranteed loans and the substitution between guaranteed and non-guaranteed credit. Martin et al. (2023) develop a model that shows that banks have the incentive to allocate guarantees to captive and risky firms and test these predictions using Spanish credit register data.

[7] The evidence in Chodorow-Reich et al. (2022) suggests that the issue of substitution may also have been relevant for the U.S. during the pandemic, as small firms that took PPP loans repaid significant non-PPP lending in the second quarter of 2020.

[8] There is evidence that the PPP scheme had significant real effects: Autor et al. (2022) and Bartik et al. (2020) document that it raised employment at eligible firms and increased firms' survival. Instead, our evidence for the euro area dovetails with that by Core and De Marco (2023) for Italy and Kozeniauskas et al. (2020) for Portugal and broadly in line with the evidence by Gourinchas et al. (2024).

2.3. Equilibrium with guaranteed loans

Suppose now that guaranteed loans of size L^G are offered after these contracts are signed. The availability of guaranteed loans induces firms and banks to change the terms of their initial contracts. In both cases, the outcome differs depending on the firm's default risk and bargaining power vis-à-vis banks in the renegotiation of the initial contracts, as shown below.

2.3.1. Loan renegotiation

Consider first the terms at which firms can renegotiate their loans with competitive lenders: upon becoming eligible for a guaranteed loan, they will want to reduce their non-guaranteed loans by the full amount of the guaranteed loans, which carries a zero interest rate, as the government absorbs its default risk. For banks to break even, the pledged repayments required from good and bad firms respectively drop to:

$$F'_{sg} = \frac{I_s - L^G}{1-p}, \quad F'_{sb} = \frac{I_s - L^G}{1-p-\Delta}. \tag{5}$$

As a result, the average interest rates pledged by the two firms on their overall (i.e., guaranteed and non-guaranteed) debt respectively drop to $\frac{p}{1-p} \cdot \frac{I_s - L^G}{I_s}$ and $\frac{p+\Delta}{1-p-\Delta} \cdot \frac{I_s - L^G}{I_s}$. The two firms' expected payoffs respectively rise to

$$\Pi'_{sg} = (1-p)(Y - F'_{sg} - L^G) = (1-p)Y - I_s + pL^G = \Pi_{sg} + pL^G, \tag{6}$$

$$\Pi'_{sb} = (1-p-\Delta)(Y - F'_{sb} - L^G) = (1-p-\Delta)Y - I_s + (p+\Delta)L^G = \Pi_{sb} + (p+\Delta)L^G, \tag{7}$$

by substituting for F'_{sg} and F'_{sb} from (5) in the second step of expressions (6) and (7). Hence, the guarantee benefits both firms by reducing their credit costs and shifting their default risk to the government. Bad firms benefit more than good ones: the former gains $(p+\Delta)L^G$, the latter pL^G.

However, the loan guarantee also enables good firms to upgrade their investment from a small to a large project by complementing the guaranteed loan L^G with a non-guaranteed loan $I_\ell - L^G$, which may be smaller or larger than the pre-existing loan I_s depending on whether the guaranteed loan exceeds the cost difference between the two projects ($L^G > I_\ell - I_s$) or not ($L^G < I_\ell - I_s$). In the first case, the guaranteed loan would still trigger credit substitution, but less than for its full amount, because $L^G - (I_\ell - I_s) \in (0, L^G)$. Hence, substitution would be *less* than for bad firms undertaking small projects, for which substitution is complete. In the second case, the guaranteed loan would generate *additional* lending $I_\ell - I_s - L^G > 0$ rather than credit substitution: banks are willing to relax good firms' borrowing constraints by funding negative-NPV projects (at least based on their pledgeable cash flow) because guaranteed loans enable them to shift credit risk to the government.

From the banks' zero-profit condition, the pledged repayment on these loans is

$$F_{\ell g} = \frac{I_\ell - L^G}{1-p}, \tag{8}$$

so that the expected payoff for good firms becomes

$$\Pi_{\ell g} = B + (1-p)(X - F_{\ell g} - L^G) = B + (1-p)X - I_\ell + pL^G > \Pi_{sg} = (1-p)Y - I_s + pL^G. \tag{9}$$

This inequality, which follows from assumption (1), indicates that good firms prefer this loan contract to one funding a small project. The repayment (8) entails that the average rate pledged by good firms on their overall debt drops to $\frac{p}{1-p} \cdot \frac{I_\ell}{I_\ell - L^G}$, down from $\frac{p}{1-p}$ in the absence of guaranteed loans. Interestingly, the implied drop in their cost of credit, $\frac{p}{1-p} \cdot \frac{L^G}{I_\ell}$, is less than that enjoyed by bad firms, $\frac{p+\Delta}{1-p-\Delta} \cdot \frac{L^G}{I_s}$.

Hence, if the introduction of guaranteed loans is followed by competitive renegotiation of the initial contract, it is predicted to entail complete substitution of non-guaranteed lending to riskier firms but incomplete substitution or even additional issuance of non-guaranteed lending to safer firms. In both cases, firms reap the entire benefit from public insurance of credit risk through reduced cost of credit. But riskier firms gain more than safer ones as they enjoy a more significant drop in their credit cost. Hence, the model predicts that the introduction of guaranteed loans generates changes in non-guaranteed credit and interest rates that are positively correlated across firms, being both larger for riskier firms. Otherwise stated, credit substitution should be cross-sectionally associated with credit cost reduction.

The above argument rests on firms being able to obtain credit at competitive terms from other banks and thus threaten their initial lender with early repayment of the initial loan upon securing a new non-guaranteed loan $I_s - L^G$ and a guaranteed loan L^G at competitive terms. But if this assumption does not hold, the firm has limited bargaining power in the renegotiation process, and the initial lender can capture at least part of the benefit from public insurance against default. In the limiting case in which the firm cannot contract with any bank other than its initial lender, this can make a take-it-or-leave-it offer to the firm: issue the guaranteed loan while raising the non-guaranteed loan rate enough to appropriate the entire benefit of the guarantee and leave the firm with an unchanged total cost of credit, and thus an unchanged payoff, so that the firm can be assumed to accept it. Such an offer will require bad firms to pledge a repayment

$$F''_{sb} = \frac{I_s}{1-p-\Delta} - L^G, \tag{10}$$

which exceeds the competitive repayment F'_{sb} in (5), thus increasing the interest rate on the non-guaranteed loan. So the market value of the bank's overall debt (inclusive of the guaranteed loan), D''_{sb}, increases by $(p+\Delta)L^G$ above its value under competitive renegotiation, $D'_{sb} = I_s$:

$$D''_{sb} = (1-p-\Delta)(F''_{sb} + L^G) + (p+\Delta)L^G = I_s + (p+\Delta)L^G. \tag{11}$$

A similar finding applies to the renegotiation of loans to good firms. By assumption (1), good firms prefer to undertake a large project, which they can fund by complementing the guaranteed loan L^G with a non-guaranteed loan of size $I_\ell - L^G$ rather than $I_s - L^G$. In this case too, the initial lender's bargaining power in renegotiation will enable it to benefit from the public guarantee. In the limiting case where it can make a take-it-or-leave-it offer to the firm, the bank will fully appropriate the surplus by requiring the firm to pledge a repayment $F''_{\ell g} = \frac{I_\ell}{1-p} - L^G$ on its non-guaranteed loan $I_\ell - L^G$, which exceeds the competitive repayment $F'_{\ell m}$ and implies an increase in the interest rate on the non-guaranteed loan. This allows the bank to appropriate the entire expected value of the guarantee, pL^G, as the value of the firm's overall debt becomes

$$D''_{\ell g} = (1-p)(F_{\ell g} + L^G) + pL^G = I_\ell + pL^G, \tag{12}$$

as opposed to $D'_{\ell g} = I_\ell$ under competitive renegotiation. Hence, also in this case, the banks' bargaining power results in a higher cost of non-guaranteed credit (though smaller than for bad firms) and an unchanged total cost of credit. Symmetrically, the firm's expected payoff drops below its value $\Pi_{\ell g}$ under competition in (9) by the expected value of the loan guarantee, pL^G.

In general, the renegotiation between the bank and the firm can be expected to lead to an intermediate outcome between the two polar cases in which the surplus created by the loan guarantee is appropriated entirely by either firms or banks. The new conditions of non-guaranteed loans will depend on the distribution of the bargaining power between the firm and its initial lender.

3. Institutional framework

As a preliminary step to our empirical analysis, this section provides information on the design of EU loan guarantee programs enacted in the wake of the COVID-19 outbreak, as the extent to which banks engage in credit substitution may be affected by the eligibility rules that determine the allocation of credit guarantees across firms. As we shall see, EU loan guarantee programs were structured so as to exclude firms whose viability could not benefit from such guarantees, namely, both those already close to insolvency before the outbreak of the pandemic, and those that were not negatively affected by the pandemic. Hence, the eligibility rules were designed so as to exclude from loan guarantee programs both the lower and the upper tail of the firm distribution by credit risk.

The design of loan guarantee schemes in European Union (EU) countries shares several common features defined by the EU Commission Regulation No. 651/2014, although national rules determine some of their details. Not all firms could benefit from loan guarantee programs in the EU. The Communication of the EU Commission about State aid during the pandemic (2020/C 91 I/01) stated: "The guarantee may be granted to undertakings that were not in difficulty ... on 31 December 2019", thus excluding firms already close to default before the start of the pandemic. The Commission's definition of an "undertaking in difficulty" is one for which at least one of the following circumstances occurs:

(a) limited liability companies (other than SMEs that existed for less than three years), where more than half of their subscribed share capital has disappeared as a result of accumulated losses,
(b) companies where at least some members have unlimited liability for the debt of the company (other than an SME that existed for less than three years), where more than half of their capital as shown in the company accounts has disappeared as a result of accumulated losses,
(c) firms subject to collective insolvency proceedings or fulfilling the criteria for being placed in collective insolvency proceedings at the request of their creditors,
(d) firms that received rescue aid and have not yet reimbursed the loan or terminated the guarantee, or received restructuring aid and are still subject to a restructuring plan,
(e) large firms, whose book debt to equity ratio exceeded 7.5 and EBITDA interest coverage ratio was below 1 for the previous two years.

At the same time, the EU Commission required aid to be targeted to firms "that faced difficulties or entered in difficulty thereafter as a result of the COVID-19 outbreak", hence excluding those firms unaffected by the pandemic or even benefiting from it.

Regarding credit substitution, national regulators appeared to tolerate it at least to some degree. For instance, French regulation subjected the guarantee to the bank evidencing that the loan granted led to an "increase in the bank's commitments to the borrower compared to commitments that existed as at 16 March 2020". In Italy, loans guaranteed by Fondo Nazionale di Garanzia and designed for refinancing of existing loans were required to involve at least 25% of new lending. The media also appeared acutely aware of the risk that loan guarantee programs might benefit banks more than the firms hit by the pandemic.[9]

The EU Commission set minimum guarantee premia increasing in maturity and more stringent for large enterprises than for SMEs, as well as a ceiling of 6 years on the maturity of all loans. It also mandated limits to the size of guaranteed loans: these could not exceed twice the annual wage bill of the beneficiary for 2019, or 25% of total turnover of the beneficiary in 2019. Interestingly, it designed the guarantees so as to leave banks with enough "skin in the game" to remain sensitive to firms' creditworthiness when granting guaranteed loans: the guarantee could not exceed 90% of the loan principal if losses are sustained *pari passu* by the bank and the state, or 35% of the loan principal if the State is junior to the bank.

National governments introduced some differences in their respective programs: while they all designed schemes in which loans' guaranteed fraction decreased with firm size (hence, more generous with SMEs than with large firms), different governments chose different schedules for the relationship between guaranteed loan fraction and firm size, as shown in Table A1. The Italian and German governments even provided 100% guaranteed loans: in the case of Italy, this applied to all loans up to €30,000 given to small firms, and in the case of Germany to firms whose loans were issued under the KfW-Schnellkredit program. But, as shown by the table, for most loans the guaranteed fraction ranges between 70% and 90%, with lower percentages applying to larger firms. The table also reveals that the Italian, German and Spanish schemes allowed public guarantees even for loans exceeding the 6-year maturity limit prescribed by the EU Commission's guidelines.

4. Data

We draw loan-level information obtained from AnaCredit, a proprietary and confidential database of the ECB and the national central banks of euro-area countries (the Eurosystem). AnaCredit is a very granular (transaction-level) database that reports 94 loan-level attributes on a monthly frequency in a harmonized way across all euro area countries, with a common reporting threshold for company loans at €25,000. This database enhances the information obtained from national credit registers already present in several euro-area countries, as the common threshold ensures that cross-country studies, like ours, are not affected by sample selection bias possibly stemming from the different reporting threshold of the national credit registers. For example, while there is no threshold for credit exposure in Spain (any credit exposure is reported), the German credit register has a 1 million euro threshold.[10]

AnaCredit covers a comprehensive set of credit instruments: term loans, non-revolving credit lines, financial leases, trade receivables and revolving credit. Both the amount already drawn under a granted facility and the undrawn part are reported: in our analysis we consider their sum, i.e., the total commitment of the bank to the debtor with respect to an instrument. AnaCredit also reports interest rates and maturities at loan level.

Importantly for our analysis, among the attributes of each loan, there is extensive information on the protection securing the bank's credit exposure, including financial guarantees provided by government entities.[11] While in some countries special identifiers were introduced to mark guarantees provided under specific COVID-19-related schemes, these are not consistently available for all the four countries considered in our sample, so we consider all guarantees provided by government entities. As a sanity check, we compare AnaCredit data for Italy with the publicly available list of government-guaranteed loans from the Italian Fondo Nazionale di Garanzia (FNG). Applying the AnaCredit filter of loans above the 25,000 euro threshold to the FNG

[9] Also in U.S. loan guarantee programs, credit substitution is tolerated according to the SBA 7(a) guidelines, which state that "7(a) loans can be used for: Refinancing current business debt" (https://www.sba.gov/partners/lenders/7a-loan-program/terms-conditions-eligibility).

[10] The reporting thresholds for the national credit register in France and Italy are €25,000 and €30,000, respectively.

[11] The database also records guarantees provided by special entities including Instituto de Crédito Oficial in Spain, Kreditanstalt für Wiederaufbau in Germany, Ministere de l'Action et des Comptes Publics in France.

data, we find a very similar number of firms receiving guaranteed credit in both databases (around 358,000 firms).[12]

We supplement these data with bank balance sheet information drawn from the ECB supervisory data to measure banks' solvency (i.e., their capital ratio and fraction of non-performing loans), liquidity (coverage ratio), and size (total assets) as of December 2019.

Our sample from AnaCredit contains a total of 2,527,108 firms present in the database before the inception of pandemic-related loan guarantee programs: 1,140,911 from France, 371,036 from Germany, 608,672 from Italy and 406,489 from Spain. These firms borrow from 838 banks in Germany, 106 in Spain, 104 in France and 158 in Italy. The subsample of firms that are recorded to have received guaranteed credit between March and August 2020 includes 500,013 firms. For about 13% of these firms, the guaranteed loan was issued by a new bank, namely, one not lending to the firm before the inception of the program. Recall that, while guaranteed credit can be of any amount, AnaCredit's €25,000 reporting threshold implies that many micro firms that likely obtained credit below this threshold do not appear in the credit register. This could be one reason why only about one fifth of firms in AnaCredit are seen to obtain guaranteed credit.

Our initial sample of 2,527,108 firms excludes data for 227,738 firms that received guaranteed loans by August 2020 but for which no loan data were recorded in AnaCredit before the inception of the loan guarantee program in February 2020. For these firms, we cannot estimate our measure of credit substitution. (To be precise, we do not know whether these firms did not receive any credit as of February 2020 or received loans below the Anacredit reporting threshold of €25,000.)

As shown in Figs. 2 and 3, most guaranteed credit was granted in Spain and Italy, with France in third place and the smallest amount in Germany. The two figures also confirm that, as seen above, the loan guarantee schemes of all four countries were designed to channel funds preferentially to SMEs, which received around 85% of the credit in each of the four countries (Fig. 2). The prevalence of small firms is even more extreme in terms of their number, especially in Italy (Fig. 3). In terms of average guaranteed loan size, German firms received the largest loans, and Italian firms received the smallest, with guaranteed loan sizes in France and Spain being in the middle, as shown by Fig. 4.

Table 1 reports the descriptive statistics for the variables used in our analysis. Panel (a) shows firm-level statistics for the full sample, Panel (b) shows those for the subsample of firms that received guaranteed loans, Panel (c) presents bank-firm-level statistics for the subsample of firms that featured multiple bank relationships and received a guaranteed loan, Panel (d) reports loan-level statistics for guaranteed loans.

Firms that received guaranteed loans are not a random sample, but one whose composition depends on eligibility guidelines. This can be seen by comparing the statistics in Panel (a), which are for the entire sample of 2,527,108 firms, with those in Panel (b), which refer to the 500,013 firms that received guaranteed credit. On average, firms that obtained guaranteed loans were those in industries more affected by the pandemic shock but ex-ante less risky than those in the whole sample. These differences are confirmed by the firm-level regressions that investigate how the allocation of guaranteed loans correlates with firm characteristics: the estimates shown in Table A2 show that the likelihood of receiving guaranteed credit was greater for smaller and less risky firms, and for those in industries more affected by the pandemic shock, even after including country-level fixed effects to control for cross-country differences. Hence, consistent with the EU Commission eligibility guidelines, guaranteed loans were neither given indiscriminately nor adversely selected: they were targeted toward adversely hit and small firms that were not ex-ante riskier, as eligibility criteria leveraged banks' information to screen out firms already close to distress before the pandemic.

Panel (d) of Table 1 reports loan-level statistics for guaranteed loans. The average percentage of the government guarantee is around 80 per cent, the average interest rate is 2 per cent and the average maturity is 33 months. The characteristics of the guaranteed loans differed across the four countries, consistently with the rules reported in Table A1. The Appendix reports country-level descriptive statistics (in Tables A10, A11, A12 and A13) that, broadly speaking, confirm for each country the firm-level observations discussed above.

5. Empirical methodology

The central question addressed in this paper is whether banks extending guaranteed loans reduce their non-guaranteed exposures toward them. Identifying credit substitution is challenging. The issue is the classical counterfactual problem: whether a firm that received a guaranteed loan would have faced a credit cut anyway. This is an important concern, especially for firms that were already weak before the pandemic and/or were severely hit by the shock: if they had not received guaranteed loans, presumably these firms would have been more likely than others to face a credit cut, and even loan foreclosure. Indeed, as mentioned above, the evidence reported in Table A2 indicates that government guarantees were not randomly allocated across euro-area firms but carefully targeted to specific firm classes. We are able to address this selection issue owing to the granularity of our data, which allows us to exploit within-firm variation. As data are at bank-firm level, for firms borrowing from multiple banks we can exploit variation in lending across banks lending to the *same* firm: specifically, we can compare the change in non-guaranteed exposures between banks extending guaranteed loans and other banks lending to a given firm, along the lines of Khwaja and Mian (2008).

Specifically, we define credit substitution as the negative of change in non-guaranteed credit ($NGC_{i,j}$) extended to firm i by bank j, divided by total initial credit granted to firm i in February 2020 by all its lenders ($TC_{i,t-1}$):

$$s_{i,j} = -\frac{NGC_{i,j,t} - NGC_{i,j,t-1}}{TC_{i,t-1}},$$

where t is (end of) August 2020 and $t-1$ is (end of) February 2020. The reason why the change in non-guaranteed credit ($NGC_{i,j}$) is normalized by total initial credit ($TC_{i,t-1}$) instead of bank j's initial credit ($TC_{i,j,t-1}$) is that this enables us to retain in the sample firms whose guaranteed loan is issued by a new bank, namely, one that was not lending to the firm before the inception of the program.[13] We estimate a specification at the bank-firm level, where variables vary across the lending relationships that a given firm i has with its banks $j = 1, \ldots, J$:

$$s_{ij} = \beta_1 L^G_{ij} + \beta_2 L^G_{ij} \times \Delta VA_i + \beta_3 L^G_{ij} \times Size_i + \beta_4 L^G_{ij} \times Risk_i + \beta_5 L^G_{ij} \times BSize_j + \beta_6 L^G_{ij} \times Liq_j + \beta_7 L^G_{ij} \times Cap_j + \beta_8 L^G_{ij} \times NPL_j + \gamma_i + \eta_j + \varepsilon_{ij} \quad (13)$$

For banks that do not grant guaranteed credit to firm i, the numerator of s_{ij} coincides with the change in their credit to the firm. Hence, if bank j grants guaranteed credit to firm i, s_{ij} measures bank j's substitution, while for other banks it measures the change in their credit to firm i. The main regressor, L^G_{ij}, is the amount of the government-guaranteed loan received by the firm i, divided by its total outstanding credit as of February 2020. The coefficient β_1 measures the magnitude

[12] The number of firms that obtained a loan guarantee is smaller in our tables since we only consider firms present in the AnaCredit database in December 2019.

[13] Normalizing by $TC_{i,j,t-1}$ would mechanically remove the observations related to these firms from the sample, as the denominator would equal zero if bank j did not lend to the firm at $t-1$.

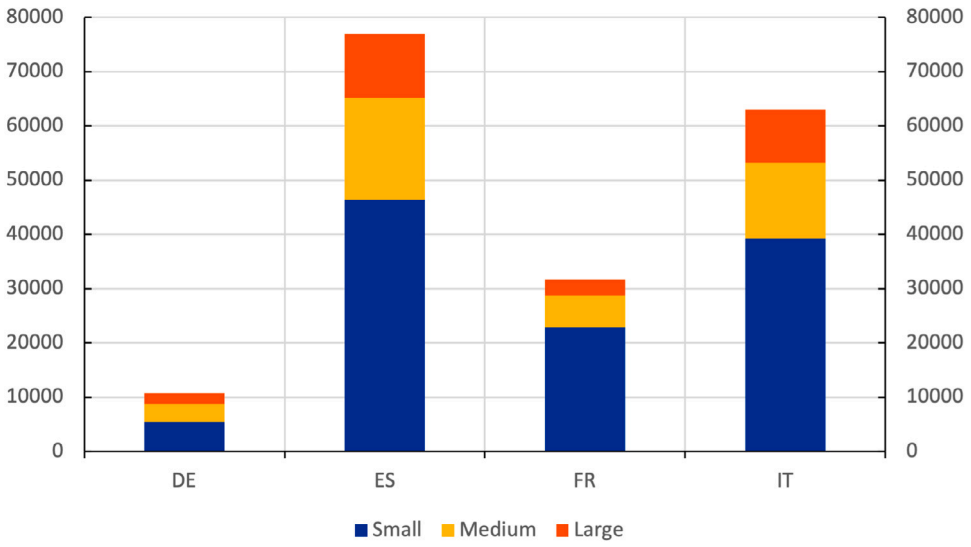


Fig. 2. Amount of guaranteed loans by firm size
The figure shows the amount of guaranteed loans (in million euros) issued to firms in different size classes based on their employment (small firms being those with less than 50 employees, medium firms those with 50 to 250 employees, and large firms as those with more than 250 employees). The sample includes firms present in the AnaCredit database as of December 2019 and considers guaranteed loans issued between March and August 2020.

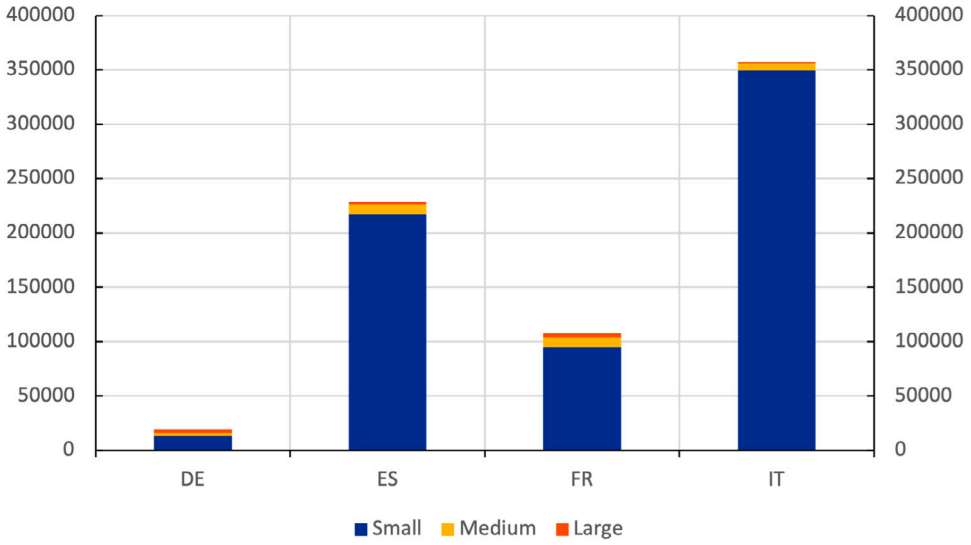


Fig. 3. Number of guaranteed loans by firm size
The figure shows the number of guaranteed loans issued to firms in different size classes to their employment size (small firms being those with less than 50 employees, medium firms those with 50 to 250 employees, large firms as those with more than 250 employees). The sample includes firms recorded in the AnaCredit database as of December 2019 and considers guaranteed loans issued between March and August 2020.

of bank j's substitution: it represents the change in the amount of non-guaranteed lending per euro of guaranteed lending granted. Since the regression includes firm fixed effects, γ_i, it exploits the variation in lending behavior across banks lending to the same firm. The other coefficients measure whether the magnitude of bank j's substitution differs depending on firm, bank, and firm-bank characteristics.

The firm characteristics that we consider are: (a) the change in the value added of firm i's industry and country during the sample period (ΔVA_i), (b) the log of firm i's total outstanding bank loans as of December 2019 ($Size_i$),[14] and (c) the firm's fraction of credit in arrears as of December 2019 ($Risk_i$), as a measure of the firm's ex-ante credit risk. The interaction of the loan guarantee variable L_{ij} with the change in sectoral value added in specification (13) is meant to capture that credit substitution may have been affected by the extent to which the firm was exposed to the recessionary effects of the pandemic. To reduce endogeneity concerns, the change of value added is not measured at the firm level, but at the country-industry level, which also takes into account that the pandemic may have differently affected the same industry in different countries. The interactions with firm size and risk are intended to allow for different degrees of credit substitution across firms, owing to banks' lower willingness to grant credit to smaller and riskier firms, which are typically subject to tighter financial constraints (Beck et al., 2005), especially at times of economic stress.

The regressions also include four bank characteristics measured before the pandemic to investigate whether bank size and balance sheet

[14] Owing to incomplete reporting of other metrics of firm size in the Anacredit database (such as total assets or number of employees), we use total bank loans as a proxy.

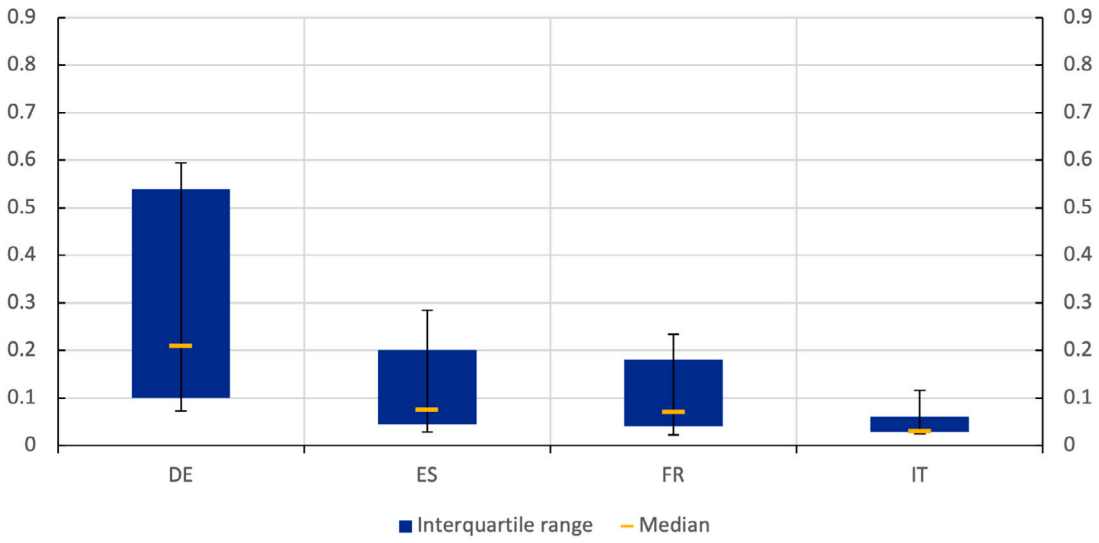


Fig. 4. Distribution of guaranteed loan size
The figure reports the distribution of the size of guaranteed loans in million euro issued in different countries. We report the median, the interquartile range and the 16th and 84th percentile. The sample includes firms recorded in the Anacredit database as of December 2019 and considers guaranteed loans issued between March and August 2020.

strength mattered for credit substitution. Our specification includes three measures of the balance sheet strength of banks lending to firm i: liquidity (Liq_j), capitalization (Cap_j), and non-performing loans as a fraction of total loans (NPL_j). Finally, some specifications also include controls at bank-firm level: the credit granted by bank j scaled by the total credit owed by the firm to all banks (as a proxy for relationship banking), the ratio between the drawn amount and credit granted, and the residual maturity of the loans issued by a bank to a given firm. All bank and bank-firm variables are measured as of December 2019. The specification also includes bank fixed effects (η_j) to control for unobserved bank heterogeneity in the provision of credit during the pandemic. Standard errors are double-clustered at the bank and firm level. We also estimate similar specifications where the dependent variables are interest rates and loan maturities.

While the main analysis is performed at the bank-firm level, thus exploiting within-firm variation, we also measure credit substitution at the firm level, to investigate whether and how guaranteed lending affected the overall non-guaranteed lending to firms. The analysis focuses only on firms that received loan guarantees to minimize the impact of selection due to eligibility criteria. We measure firm-level credit substitution by the drop in non-guaranteed loans to a given firm in response to an extra euro of guaranteed loans, i.e., the negative of the change in non-guaranteed credit (NGC_i) to firm i, scaled by its initial total credit (TC_i):

$$s_i = -\frac{NGC_{i,t} - NGC_{i,t-1}}{TC_{i,t-1}},$$

where t is (end of) August 2020 and $t-1$ is (end of) February 2020. Substitution occurs if non-guaranteed credit declines upon the firm receiving a guaranteed loan, i.e., if $s_i > 0$. However, in principle, non-guaranteed credit may increase, in which case the measure s_i would turn negative. Fig. 5 shows how the variable s_i is distributed across firms for the whole sample and separately for each of the four countries: it is positive for most firms, its median being positive in all countries, and larger in Italy and Spain than in France and Germany. In Italy and Spain, almost the whole distribution is in positive territory. Hence, this simple unconditional statistic suggests that substitution was larger in Italy and Spain than in France and Germany. However, this result may reflect cross-country differences in firm characteristics, as well as in the magnitude of the shock hitting them.

To account for such heterogeneity, we investigate how substitution is related to the size of the guaranteed loan scaled by total initial credit, L_i^G, and to its interactions with firm and bank characteristics, for the subsample of firms that received a guaranteed loan:

$$s_i = \alpha + \beta_1 L_i^G + \beta_2 L_i^G \times \Delta VA_i + \beta_3 L_i^G \times Size_i + \beta_4 L_i^G \times Risk_i + \beta_5 L_i^G \times BSize_i + \beta_6 L_i^G \times Liq_i + \beta_7 L_i^G \times Cap_i + \beta_8 L_i^G \times NPL_i + \gamma_c + \varepsilon_i \quad (14)$$

While firm variables are defined as in Eq. (1), bank variables are weighted averages of the corresponding values for the banks lending to firm i, with weights equal to their shares in the firm's total credit as of December 2019. In estimating this specification, errors are clustered at the level of the relevant firm's main bank.

6. Results

In this section we address the main issue of the paper, by assessing the extent to which guaranteed loans resulted in credit substitution, and investigating how it correlates with the firm and bank characteristics highlighted as potentially relevant in the model presented in Section 2. To identify the effect of the loan guarantee program on the availability of non-guaranteed credit to firms that received guarantees, in Section 6.1 the analysis is carried out at within-firm level, by comparing credit granted by lenders providing a guaranteed loan with that issued by other lenders catering to the same firm. For robustness, we inquire whether the results hold if we exclude non-guaranteed loans that mature around the inception of guaranteed loan programs, as these would mechanically generate the appearance of credit substitution. We also test whether the results differ between term loans that cannot be unilaterally revoked by the bank and callable credit lines.

In Section 6.2, we address further potential selection issues. First, we try to sign the possible selection bias arising from the non-random choice of the lender providing the guaranteed loan among the firm's lenders. Second, we check whether our results hold in the full population of firms, including single-bank firms.

Section 6.3 investigates how the provision of guaranteed loans correlates with changes in the interest rate and the maturity offered by the bank granting the guaranteed loan, and whether these changes are correlated with credit substitution as predicted by our model.

Table 1
Descriptive statistics.

Variable	Observations	Mean	Std. dev.	p25	p50	p75
Panel (a): Firm-level statistics for full sample						
G_i	2,527,108	0.22	0.41	0.00	0.00	0.00
L_i^G	2,527,108	0.15	0.40	0.00	0.00	0.00
Industry VA Growth	2,527,108	−0.18	0.13	−0.28	−0.21	−0.04
Firm Size (ln)	2,527,108	−1.84	1.39	−2.80	−1.96	−1.02
Firm Risk	2,527,108	0.05	0.20	0.00	0.00	0.00
Bank Assets (ln)	2,492,992	11.02	1.87	9.75	10.68	12.70
Bank Liquidity	1,970,077	1.59	0.44	1.37	1.45	1.64
Bank Capital	2,461,231	0.17	0.04	0.14	0.16	0.19
Bank NPL	2,478,597	0.04	0.02	0.02	0.03	0.04
Variable	Observations	Mean	Std. Dev.	p25	p50	p75
Panel (b): Firm-level statistics for sample of firms receiving a guaranteed loan						
s_i	500,013	0.25	0.53	0.01	0.20	0.59
L_i^G	500,013	0.71	0.61	0.26	0.55	0.97
Guar. Amount	474,103	0.59	0.53	0.21	0.44	0.80
Industry VA Growth	500,013	−0.27	0.12	−0.30	−0.28	−0.25
Firm Size (ln)	500,013	−1.71	1.44	−2.77	−1.87	−0.79
Firm Risk	500,013	0.02	0.10	0.00	0.00	0.00
Bank assets (ln)	495,051	11.66	1.61	10.51	12.09	12.92
Bank Liquidity	462,323	1.63	0.40	1.37	1.52	1.72
Bank Capital	488,049	0.16	0.03	0.14	0.16	0.18
Bank NPL	493,273	0.05	0.03	0.03	0.04	0.07
Panel (c): Bank-firm-level statistics for sample of firms with multiple bank relationships receiving a guaranteed loan						
s_{ij}	817,808	0.04	0.30	0.00	0.01	0.11
G_{ij}	817,808	0.51	0.50	0.00	1.00	1.00
L_{ij}^G	817,808	0.20	0.35	0.00	0.00	0.25
Industry VA Growth	813,774	−0.28	0.12	−0.30	−0.27	−0.25
Firm Size (ln)	806,194	−0.42	1.56	−1.56	−0.53	0.62
Firm Risk	812,803	0.02	0.09	0.00	0.00	0.00
Bank assets (ln)	813,781	11.32	1.76	10.10	11.74	12.92
Bank Liquidity	670,763	1.70	0.48	1.32	1.54	1.79
Bank Capital	791,748	0.15	0.03	0.14	0.14	0.17
Bank NPL	803,958	0.05	0.03	0.03	0.04	0.06
Share of gran.	817,808	0.32	0.24	0.12	0.27	0.48
Drawn/Granted	808,512	0.83	0.29	0.71	0.98	1.00
Residual Maturity (ln)	737,745	3.11	1.42	2.60	3.53	3.98
Residual Maturity	816,112	34.61	33.04	8.07	30.90	51.03
Panel (d): Loan-level statistics for guaranteed loans						
Guar. Percent	580,779	0.81	0.21	0.80	0.80	1.00
Interest Rate	595,288	1.98	1.34	1.21	1.65	2.63
Maturity	578,893	33	26.20	26	36	60

This table reports the descriptive statistics for the variables used in our analysis. Panel (a) reports the statistics at the firm level for the full sample, Panel (b) reports them for the sample of firms receiving a guaranteed loan, Panel (c) reports bank-firm-level statistics for the sample of firms with multiple bank relationships receiving a guaranteed loan, and Panel (d) reports loan-level statistics for guaranteed loans. The dummy G_i equal to 1 if firm i receives a government guaranteed loan between March 2020 and August 2020, and 0 otherwise (at the bank-firm level, the dummy G_{ij} is equal to 1 if bank j gives a government guaranteed loan to firm i between March 2020 and August 2020, and 0 otherwise); L_i^G is measured at the firm level and is defined as the amount of the government guaranteed loan received by the firm, divided by total credit in February 2020, while L_{ij}^G is defined at the bank-firm level and is defined as the amount of the government guaranteed loan received by the firm i by bank j, divided by total initial credit granted to firm i in February 2020; Industry VA Growth is defined as the percentage change in Valued Added in the relevant industrial sector in each country between February 2020 and August 2020; Firm Size is proxied by the log of firm total debt; Firm Risk is proxied by the share of loans in arrears out of total loans; Bank Assets is defined as the log of total bank assets; Bank Liquidity is defined as the bank Liquidity Coverage Ratio; Bank Capital is defined as the Core Tier 1 Ratio; Bank NPL is defined as the share of NPL loans out of its total loans; Credit substitution (s_i) is measured at firm level and is defined as the negative of the change in non-guaranteed credit received by firm i between February 2020 and August 2020, divided by its total credit as of February 2020, while s_{ij} is measured at the bank-firm level and is defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020); Guar. Amount is defined as the amount of the government-guaranteed credit multiplied by the percentage of guarantee received by the firm, divided by total credit in February 2020; Share of granted is defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted is defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual Maturity is defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i; Guar. Percent is the percentage of the guaranteed loan covered by the public guarantee. Interest Rate is the rate charged on the guaranteed loan; Maturity is the log of the number of months of the guaranteed loan's maturity. In Panels (a) and (b) bank-level variables are measured as weighted averages of the values for the banks lending to a given firm, where the weights are the shares of the bank exposure toward the firm in the firm's total bank exposure as of December 2019. All variables, apart from Guarantee, Industry VA Growth, Interest rate, and Maturity are measured as of December 2019.

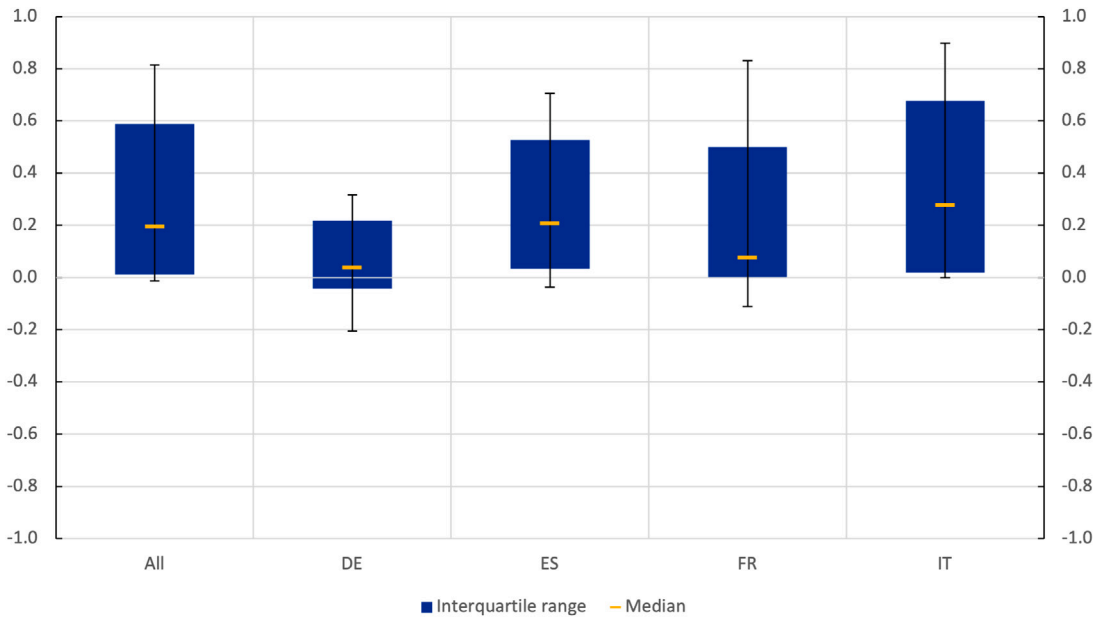


Fig. 5. Distribution of firm-level credit substitution
The figure shows the country-level distribution of the firm-level credit substitution defined as the negative of the change in non-guaranteed credit between February 2020 and August 2020, divided by total credit in February 2020 (s_i for firm i). Each box plot displays the median, the interquartile range and the 16th and 84th percentile. The sample includes all firms in the AnaCredit database as of December 2019 that received a guaranteed loan between March and August 2020.

Finally, in Section 6.4, we repeat the estimation on firm-level data, including firms borrowing from a single borrower in the sample, to obtain a firm-level estimate of credit substitution.

6.1. Within-firm estimates

Fig. 6 shows the average degree of credit substitution for the subsample of firms that received a guaranteed loan and had multiple bank relationships. Panel (a) of the figure plots credit substitution $s_{i,j}$ for bank-firm relationships with a guaranteed loan and those without. The figure shows that, for the whole sample and for each of the four countries, the banks granting guaranteed loans featured much greater credit substitution than other banks lending to the same firm: indeed, while the former reduced their non-guaranteed exposures upon granting guaranteed loans, the latter on average increased their non-guaranteed exposures, and thus mitigated total credit substitution. On average, for the four largest euro area countries, banks that granted guaranteed loans reduced their non-guaranteed credit by 12%, while banks that did not provide guaranteed loans increased their exposure by 5% during the sample period. Panel (b) of Fig. 6 displays the same statistics for a subsample that excludes firms that obtained a guaranteed loan from a new bank, namely, one not lending to the firm before the inception of guaranteed loan programs. These are cases where credit substitution cannot be positive by definition, as the bank had no prior exposure to the relevant firm. Owing to the exclusion of these firms, in this subsample the banks granting guaranteed loans reduced non-guaranteed credit by 16% on average.

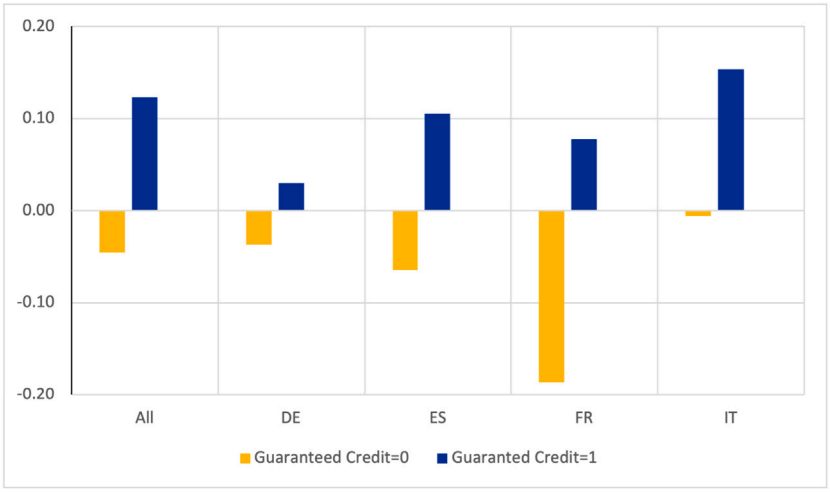
Table 2 reports estimates of substitution regressions at the within-firm level. The specification in the first column includes only firm and bank fixed effects. Other columns show estimates based on specification (13) in Section 5. The specification in the last column also controls for variables that vary at the bank-firm level (share of granted credit, ratio between drawn and granted credit, and residual maturity). In the first column, a 1-euro increase in guaranteed lending is associated with a reduction of 29 cents in non-guaranteed credit. In the other columns, whose specifications include interaction terms, substitution ranges between 28 and 32 cents. The coefficient is statistically significant at the 1% level in all specifications.

The coefficient estimates of interactions between L_{ij}^G and the three firm-level variables, i.e., industry-level value-added growth, size, and risk, reveal that substitution is larger for more fragile firms, namely, smaller and riskier ones, and those in sectors and countries more affected by the pandemic. This evidence is consistent with the prediction of the model in Section 2 that riskier firms should face more credit substitution upon getting a guaranteed loan. Moreover, substitution is estimated to be larger for banks with more liquidity, more capital and fewer NPLs before the pandemic. Finally, upon interacting L_{ij}^G with bank-firm level variables, we find that substitution is higher the longer the residual maturity in the bank-firm relationship, the smaller the size of the exposure of the issuing bank relative to the other banks lending to the same firm at the beginning of the period, and the lower the ratio between drawn and granted credit.

To take into account that guaranteed loans differ not only in size but also in their fractional guarantee, as explained in Section 3, Table A3 presents results where the guaranteed loan size, L_{ij}^G, is replaced by the guaranteed amount, which is equal to L_{ij}^G multiplied by the fractional guarantee. The results are very similar but, of course, the size of the substitution coefficient is larger, as the magnitude of the explanatory variable is smaller than in the baseline regressions of Table 2.

In the sample of firms obtaining guaranteed credit in the period under analysis, around 13% received it from a bank that was not lending to them before the program's inception. In other words, these firms established a new banking relationship through the guaranteed loan program. Consistently with the statistics reported in Panel (b) of Fig. 6, removing these bank-firm relationships from the sample raises the estimated substitution. The estimate in the first column of Table A4 of the Appendix shows that a 1-euro increase in guaranteed lending is associated with a reduction of 43 cents in non-guaranteed credit compared to 29 cents in the baseline regression shown in Table 2. Hence, the presence of banks offering guaranteed loans with no prior exposure to the firm reduces average credit substitution by about one-third.

(a) Full sample



(b) Subsample excluding firms receiving guaranteed credit from banks not formerly lending to them

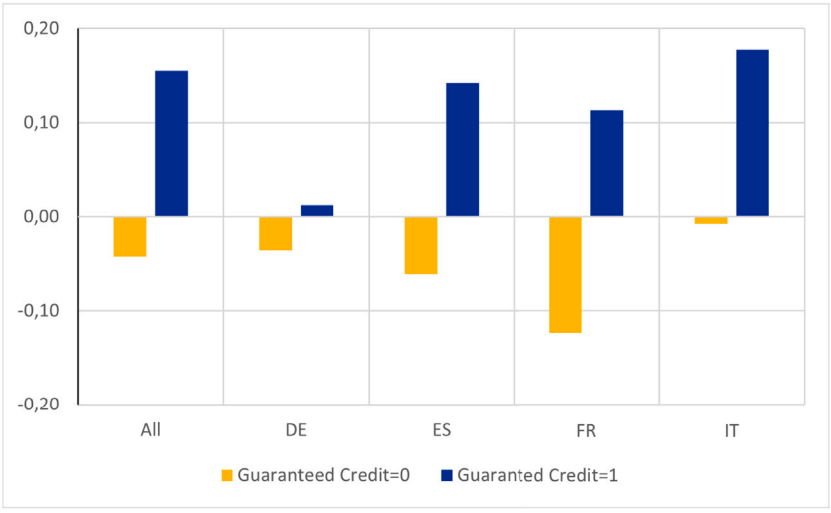


Fig. 6. Credit substitution for bank-firm relationships with and without guaranteed credit
The figure reports bank-firm level descriptive statistics of the variable s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The figure shows the average value of s_{ij} for bank-firm relationships where bank j issues a government-guaranteed loan to firm i between March 2020 and August 2020, and 0 otherwise. Panel (a) is based on the sample of all firms in the Anacredit database as of December 2019 that received a government-guaranteed loan and had multiple bank relationships. Panel (b) is based on a subsample that excludes firms whose guaranteed credit was issued by new banks, i.e., banks not lending to them as of February 2020..

Table A5 of the Appendix presents the estimates for the most complete specification (shown in column 5 of Table 2) separately for each of the four countries. The results are similar across the four countries, which is far from obvious in light of the cross-country differences in the designs of the programs, the magnitude of the pandemic shocks, the composition of firm populations, and the national banking systems. The coefficient of the L_{ij}^{G} variable is positive and highly statistically significant in all four countries. Although the magnitude of substitution varies across countries, being largest in Italy (31 cents per euro of guaranteed lending) and smallest in Germany (14 cents), in all four countries firm and bank characteristics appear to play a similar role in affecting the extent of substitution: this is larger for more fragile firms (i.e. smaller and riskier ones in more affected sectors and countries) and for stronger banks. In some cases, the relevant coefficients are not statistically significant but their signs are consistent across countries.

A potential concern with these results is that credit substitution by the banks providing guaranteed loans may not be due to an active decision on their part but simply to the expiration of a preexisting loan contract that is not rolled over upon issuing a guaranteed loan. The result that substitution is larger for banks whose pre-existing loans had longer residual maturity mitigates this concern, as it suggests that substitution is unlikely to stem from the expiration of pre-existing non-guaranteed credit. However, to address this concern more directly, in Table 3 we re-estimate the main specifications of Table 2 on a sample that excludes the bank-firm relationships with at least one loan maturing between March 2020 and August 2020. The results remain similar after applying this filter, which removes around 14% of the observations. For further robustness, in Table A6 of the Appendix, we instead remove all the firm-bank relationships where the weighted average residual maturity in February 2020 is below 6 months (the weights being based on the amounts of the various loan instruments

Table 2
Credit substitution: firm-bank level analysis.

Dependent variable:	Credit Substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.291***	0.322***	0.278***	0.278***	0.281***
	(0.016)	(0.018)	(0.016)	(0.017)	(0.016)
L_{ij}^G ×Industry VA Growth		−0.190**	−0.089*	−0.089*	−0.118**
		(0.076)	(0.051)	(0.048)	(0.049)
L_{ij}^G ×Firm Size		−0.036***	−0.033***	−0.033***	−0.028***
		(0.011)	(0.010)	(0.010)	(0.010)
L_{ij}^G ×Firm Risk		0.341***	0.324***	0.311***	0.132**
		(0.062)	(0.059)	(0.060)	(0.057)
L_{ij}^G ×Bank Assets		0.013**	0.002	0.004	0.009
		(0.007)	(0.007)	(0.008)	(0.007)
L_{ij}^G ×Bank Liquidity		0.001***	0.000*	0.003***	0.001***
		(0.000)	(0.000)	(0.000)	(0.000)
L_{ij}^G ×Bank Capital		0.891*		1.046**	0.927**
		(0.527)		(0.478)	(0.417)
L_{ij}^G ×Bank NPL			−2.757***	−2.953***	−3.865***
			(0.510)	(0.540)	(0.587)
L_{ij}^G ×Share of Granted					−0.518***
					(0.032)
L_{ij}^G ×Drawn/Granted					−0.229***
					(0.044)
L_{ij}^G ×Residual Maturity					0.046**
					(0.020)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R^2	0.502	0.495	0.497	0.498	0.532
N	817 808	599 302	599 360	589 477	509 100

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^G and Industry VA Growth are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

in the total credit within the bank-firm relationship): also in this case, the results remain stable. This evidence shows that our results are not mechanically driven by loans maturing immediately after the inception of the loan guarantee programs.

Next, we investigate whether credit substitution is greater for credit instruments that can be revoked unilaterally by the bank than for those that cannot. In our data, there are three main types of credit instruments: (i) term loans, (ii) non-revolving credit lines, which cannot be revoked unilaterally by the issuing bank before maturity, and (iii) revolving credit lines, which have no maturity and can be revoked by the bank at any time with no prior notice, and amount to about 7% of credit extended via the other two instruments. Hence, banks have a lower ability to renegotiate the first two types of credit instruments when issuing guaranteed loans. In Table 4, the dependent variable is re-defined as credit substitution referring to items (i) and (ii), whereas in Table 5 is defined as substitution regarding revolving credit lines. As one would expect, the callable nature of the revolving credit lines makes substitution easier than for term loans (and non-revolving credit lines): credit substitution is larger (indeed, over twice as large) for revolving credit lines than for term loans and non-revolving credit lines.[15] Using the estimates in column 5 of Tables 4 and 5, we find that for term loans and non-revolving credit lines, a 1-euro increase in guaranteed lending is associated with a reduction of 25 cents in non-guaranteed credit relative to other banks lending to the same firm, while for revolving credit lines the reduction is of about 43 cents.

6.2. Other potential selection issues

While the within-firm estimates shown in Table 2 avoid firm selection issues, they are not entirely immune from selection bias. First, banks issuing guaranteed loans are not randomly assigned across a given firm's pre-existing lenders. Second, the characteristics of firms with multiple bank relationships may differ from those of the population.

To sign the first potential selection bias, we analyze the within-firm selection of the bank granting guaranteed credit: Table 6 investigates the characteristics of the banks granting guaranteed credit within the subsample of firms with multiple banking relationships. The dependent variable is a dummy variable equal to 1 for banks granting guaranteed credit, and 0 otherwise.

The coefficient estimates of the bank size and capitalization variables in Table 6 show that the banks that provide guaranteed credit are significantly larger and more capitalized. Moreover, these banks

[15] The results of Table 4 are robust to the exclusion of maturing loans as shown in Table A7 in the Appendix (an exercise that, of course, cannot be repeated for revolving credit lines since they have no maturity). Once maturing loans are removed, the difference between the estimates of credit substitution for callable and non-callable credit instruments becomes even larger.

Table 3
Credit substitution: firm-bank level analysis, excluding relationships with maturing loans.

Dependent variable:	Credit substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.281***	0.308***	0.269***	0.259***	0.262***
	(0.015)	(0.021)	(0.019)	(0.020)	(0.019)
L_{ij}^G ×Industry VA Growth		−0.057	−0.001	−0.016	−0.020
		(0.050)	(0.044)	(0.045)	(0.036)
L_{ij}^G ×Firm Size		−0.034***	−0.032***	−0.032***	−0.028***
		(0.009)	(0.009)	(0.010)	(0.010)
L_{ij}^G ×Firm Risk		0.336***	0.309***	0.302***	0.132**
		(0.060)	(0.059)	(0.060)	(0.063)
L_{ij}^G ×Bank Assets		0.005	0.004	0.014	0.015**
		(0.010)	(0.008)	(0.011)	(0.008)
L_{ij}^G ×Bank Liquidity		0.016	0.022	0.017	0.006
		(0.048)	(0.044)	(0.046)	(0.035)
L_{ij}^G ×Bank Capital		1.045**		0.980*	0.830**
		(0.480)		(0.511)	(0.405)
L_{ij}^G ×Bank NPL			−2.773***	−3.001***	−3.872***
			(0.482)	(0.506)	(0.554)
L_{ij}^G ×Share of Granted					−0.511***
					(0.033)
L_{ij}^G ×Drawn/Granted					−0.241***
					(0.044)
L_{ij}^G ×Residual Maturity					0.045**
					(0.021)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R^2	0.503	0.494	0.497	0.498	0.531
N	705 965	510 099	500 477	491 345	417 784

This table reports bank-firm level estimates of a regression whose dependent variable is the credit substitution s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The sample excludes bank-firm relationships where there is at least one loan maturing between March 2020 and August 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm divided by the total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual Maturity is defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^G and Industry VA Growth are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

are more likely to be the relevant firms' main banks, as they feature a significantly larger share of granted credit. Thus, relationship lending appears to have made it easier for firms to access government-guaranteed credit, consistent with evidence by Li and Strahan (2021) that the bank supply of credit under the Paycheck Protection Program (PPP) was done mainly by relationship banks. The results in Table 6 help us infer the sign of the potential bias in the substitution estimates of Table 2 arising from the selection of the banks providing guaranteed credit: these "selected" banks are stronger, and more likely to engage in relationship lending with the relevant firms: hence, they are precisely the type of banks that according to the literature (Bolton et al., 2016, and Jiménez et al., 2012) should be associated with greater supply of credit during economic shocks. By extension, these banks should also be associated with lower credit substitution. Hence, if anything, the estimates in Table 2 are a lower bound on the extent of substitution by banks providing guaranteed loans.[16]

Moreover, the results in Table 6 provide further reassurance that our results are likely to stem from banks' decision to cut their non-guaranteed credit when issuing a guaranteed loan, rather than to casual concomitant expiration of pre-existing non-guaranteed loans. When residual maturity is included in the specifications of columns 4–6, its coefficient is positive and significant: the bank issuing the guaranteed loan tends to be the one whose credit to the firm had the longer residual maturity, confirming that the previous findings regarding substitution are not simply driven by expiring credit being rolled over in the form of guaranteed credit.

The regressions reported so far are estimated on the subsample that only contains firms that borrowed from multiple banks, one of which granted a guaranteed loan to them, so as to merely exploit within-firm variation in non-guaranteed credit substitution. However, this not only greatly reduces the sample, but may introduce selection biases, as the characteristics of firms with multiple-bank relationships may differ from those of the population. To address this issue, in Table 7 we estimate regressions that also include single-bank firms. Even though these regressions cannot include firm fixed effects, they include industry-location-size (ILS) fixed effects as in Degryse et al. (2019) and Acharya et al. (2019). The results show that the inclusion of single-bank

[16] Another reason why our within-firm estimates may underestimate credit substitution is that guaranteed loans may have a negative spillover effect on non-guaranteed lending by banks not issuing the guaranteed loan: these lenders may reassess the firm as being riskier because of reduced monitoring by the bank issuing the guaranteed loan, and thus reduce their non-guaranteed lending to the firm alongside with the bank issuing the guaranteed loan. If so, using these lenders as benchmarks for the credit response of the bank issuing the guaranteed loan would underestimate the magnitude of its credit substitution.

Table 4
Credit substitution: firm-bank level analysis. Term loans plus non-revolving credit lines.

Dependent variable:	Credit substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.138***	0.249***	0.212**	0.266**	0.252**
	(0.018)	(0.044)	(0.073)	(0.056)	(0.093)
L_{ij}^G ×Industry VA Growth		−1.180*	−0.695*	−0.533	−0.546
		(0.615)	(0.386)	(0.436)	(0.422)
L_{ij}^G ×Firm Size		−0.241***	−0.147**	−0.141*	−0.136*
		(0.082)	(0.069)	(0.073)	(0.075)
L_{ij}^G ×Firm Risk		3.415***	3.044***	3.083***	3.503***
		(0.620)	(0.511)	(0.510)	(0.615)
L_{ij}^G ×Bank Assets		0.412***	0.501***	0.495***	0.391***
		(0.133)	(0.118)	(0.115)	(0.116)
L_{ij}^G ×Bank Liquidity		0.992**	1.143***	0.995***	0.726**
		(0.437)	(0.329)	(0.323)	(0.311)
L_{ij}^G ×Bank Capital		5.540**		3.309	2.938
		(5.630)		(2.429)	(2.830)
L_{ij}^G ×Bank NPL			−4.601***	−4.855***	−5.101***
			(0.432)	(0.413)	(0.436)
L_{ij}^G ×Share of gran.					−2.647***
					(0.306)
L_{ij}^G ×Drawn/Granted					−0.807***
					(0.209)
L_{ij}^G ×Residual Maturity					0.021
					(0.103)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
N	335 266	239 588	241 806	237 284	213 349
r2	0.678	0.716	0.718	0.720	0.545

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. In these regressions, "credit" refers exclusively to term loans plus non-revolving credit lines. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

firms in the sample does not qualitatively affect the main coefficients of interest (both in terms of credit substitution and its interactions with firm and bank characteristics), except that the estimates of credit substitution are approximately 2 to 6% higher.

6.3. Change in interest rate and loan maturity

The model in Section 2 shows that public loan guarantees should simultaneously affect both the amount of non-guaranteed lending and the interest rate offered by the bank issuing the guaranteed loan, and that firm and bank characteristics should affect both of them. Indeed, the model predicts that the renegotiation of preexisting credit contracts between banks and firms eligible for guaranteed loans generates a positive cross-sectional correlation between credit substitution and the drop in the cost of credit, as both should be greater for riskier firms. A similar trade-off can be expected between substitution and loan maturity: the bank can lengthen the maturity of the firm's loans as another concession to clients who bring loan guarantees to the negotiation table.

To analyze how interest rates and loan maturities respond to the availability of loan guarantees in our data, and how such response varies depending on bank and firm characteristics, we adopt the same econometric specification used to assess the extent of credit substitution, namely, Eq. (13) in Section 5, replacing the dependent variable with the change in the interest rate or in the loan maturity of the relevant bank-firm relationship over the relevant period. In line with the model's setup, we consider how the bank providing the guaranteed loan modifies the interest rate and loan maturity that apply to its overall lending position to the firm, i.e., the respective weighted averages applying to all the loans and credit lines issued to the relevant firm.

In Table 8, the dependent variable is $\Delta InterestRate_{ij}$, i.e., the change between February 2020 and August 2020 in the weighted average interest rate charged by bank j to firm i. The estimates indicate that firms that receive a loan guarantee face a reduction in the interest rate charged on the loans extended to the firm by the bank issuing the guaranteed loan. The interest rate reduction amounts to 17 basis points for the average bank-firm relationship featuring a guaranteed loan relative to other banks lending to the same firm.[17]

The interest rate drop is larger for more fragile firms, i.e., smaller and riskier ones and those belonging to more affected sectors and countries. The result that riskier firms obtain a larger interest rate concession is in line with the predictions of the model presented in Section 2, which highlights that the credit risk insurance provided by the government via loan guarantees is more valuable for riskier clients, and therefore should benefit more such clients via a lower credit cost, to

[17] This number is obtained by multiplying the coefficient of L_{ij}^G (−0.426) by the average guaranteed loan scaled by the total initial credit for bank-firm relationships where there is a guaranteed loan (0.39).

Table 5
Credit substitution: firm-bank level analysis. Revolving credit lines.

Dependent variable:	Credit substitution ($s_{i,j}$)				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.426***	0.464***	0.402***	0.424***	0.427***
	(0.082)	(0.093)	(0.094)	(0.106)	(0.115)
L_{ij}^G ×Industry VA Growth		−0.808***	−0.689***	−0.634***	−0.367*
		(0.249)	(0.231)	(0.222)	(0.188)
L_{ij}^G ×Firm Size		−0.089***	−0.091***	−0.088***	−0.125***
		(0.017)	(0.015)	(0.017)	(0.030)
L_{ij}^G ×Firm Risk		0.777***	0.830***	0.840***	1.790***
		(0.219)	(0.216)	(0.217)	(0.350)
L_{ij}^G ×Bank Assets		0.142***	0.0942***	0.112***	0.111**
		(0.028)	(0.030)	(0.031)	(0.043)
L_{ij}^G ×Bank Liquidity		0.552***	0.407***	0.513***	0.471***
		(0.133)	(0.124)	(0.132)	(0.161)
L_{ij}^G ×Bank Capital		4.407***		3.930***	2.955***
		(1.085)		(1.017)	(1.081)
L_{ij}^G ×Bank NPL			−3.363***	−3.862***	−3.386***
			(1.853)	(1.863)	(2.309)
L_{ij}^G ×Share of gran.					−0.574***
					(0.212)
L_{ij}^G ×Drawn/Granted					−0.609***
					(0.222)
L_{ij}^G ×Residual Maturity					0.151**
					(0.071)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
N	428 761	326 285	327 720	323 854	256 201
r2	0.669	0.752	0.752	0.753	0.624

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. In these regressions, "credit" refers exclusively to revolving credit lines. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

the extent that lending competition forces banks to transfer this benefit to them.

The specifications in columns 5 and 6 of Table 8, which also control for characteristics of the bank-firm relationship, indicate that the interest rate cut offered by the bank issuing the guaranteed loan is larger if the bank features a high NPL/loan ratio and supplies a large fraction of the firm's total credit. These effects may be interpreted in the context of the bank-firm renegotiation when the firm becomes eligible for a guaranteed loan: the larger the firm's preexisting exposure vis-à-vis its bank and the worse the bank's preexisting loan portfolio, the greater the benefit to the bank from a loan guarantee and thus its willingness to offer an interest rate cut to the firm. This aligns with the model's prediction that banks with weaker bargaining power vis-à-vis their clients transfer to these firms a larger fraction of the benefit from loan guarantees.

Next, we investigate how the reduction in interest rate offered to firms with guaranteed loans between February and August 2020 correlates with credit substitution in the sample of bank-firm relationships. Specifically, we take to the data two predictions of our model regarding the joint response of interest rates and loan amounts to the provision of loan guarantees. First, in renegotiating pre-existing loans, banks should offer larger interest rate concessions to clients to whom they reduce non-guaranteed credit by more, i.e., for whom they engage in more credit substitution, since the model predicts that riskier clients will benefit more from the availability of loan guarantees but also face more credit substitution. Second, in such renegotiation, banks with greater bargaining power should grant a smaller interest rate concession to clients to whom they extend a guaranteed loan than banks with less bargaining power, in exchange for a given amount of credit substitution; symmetrically, clients with greater bargaining power should obtain a more favorable tradeoff between interest rate concessions and credit substitution.

We consider two characteristics of bank-firm relationships that should affect the distribution of bargaining power between banks and firms with guaranteed loans. First, we distinguish between firms with a single lender and those borrowing from multiple banks: in the former case lenders can be expected to have more bargaining power than in the latter, and therefore should grant a lower interest rate concession. Second, we split the sample between firms with at least one revolving credit line as of February 2020 and those without a revolving credit line: the former were exposed to a unilateral withdrawal or reduction of their non-guaranteed credit by the bank, while the latter were not, so that in the former case we can expect the lender to offer a smaller interest rate concession.

To test these predictions, in Table 9 we estimate regressions in which the dependent variable is the change in the interest rate charged by the bank issuing the guaranteed loan, and the explanatory variable is the degree of credit substitution received by the firm from that specific bank. To be precise, the sample includes single-bank firms and firms with multiple banking relationships. For the latter, we only include the

Table 6
Characteristics of banks providing guaranteed loans.

Dependent variable:	Guaranteed loan ($G_{i,j}$)					
	(1)	(2)	(3)	(4)	(5)	(6)
Bank Assets	0.055***	0.071***	0.059***	0.069***	0.086***	0.071***
	(0.001)	(0.023)	(0.013)	(0.012)	(0.026)	(0.014)
Bank Liquidity	0.048	0.061	0.011	0.002	0.065	0.005
	(0.282)	(0.040)	(0.030)	(0.030)	(0.043)	(0.033)
Bank Capital	3.704***		3.687***	4.281***		4.257***
	(0.616)		(0.596)	(0.784)		(0.762)
Bank NPL		−1.215	−0.788		−1.090	−0.261
		(1.620)	(1.372)		(1.953)	(1.677)
Share of Granted	0.648***	0.676***	0.651***	0.430***	0.466***	0.430***
	(0.036)	(0.044)	(0.037)	(0.038)	(0.049)	(0.038)
Drawn/Granted	0.206***	0.167***	0.207***	0.0750	0.107**	0.149***
	(0.075)	(0.062)	(0.075)	(0.059)	(0.054)	(0.059)
Residual Maturity (ln)				0.109***	0.103***	0.110***
				(0.021)	(0.026)	(0.021)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.401	0.367	0.405	0.441	0.397	0.445
N	629 964	630 117	619 623	554 537	555 058	545 767

The table reports bank-firm level estimates of a regression whose dependent variable is a dummy G_{ij} equal to 1 if bank j gives a government-guaranteed loan to firm i between March 2020 and August 2020, and 0 otherwise. The explanatory variables are bank and firm characteristics. The bank variables are: Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of total loans. The bank-firm variables are: Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual Maturity is defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables are measured as of December 2019. Standard errors, clustered at the bank level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

bank-firm relationship where the bank granted the guaranteed loan. Thus, the regressions cannot include firm fixed effects; however, they include bank fixed effects. In the specification shown in column 1 of the table, which refers to the whole sample described above, the coefficient of credit substitution is negative and precisely estimated, consistent with the model's prediction that credit substitution should be cross-sectionally correlated with lower cost of credit to firms. The estimated coefficient implies that 100% credit substitution would be associated with an interest rate reduction of 79 basis points. In columns 2 and 3, where the sample is split between firms borrowing from one bank and those with multiple bank relationships, we find that the interest rate reduction associated with 100% credit substitution is significantly smaller for the former (28 bp) than for the latter (109 bp): conditional on the same amount of credit substitution, firms with multiple lenders benefited from a much sharper reduction in the cost of credit. Insofar as a firm with multiple bank relationships faces more competitive credit supply than one with an exclusive relationship, this result dovetails with the model's prediction that firms that can renegotiate non-guaranteed credit on a more competitive market obtain a greater interest rate concession upon receiving a guaranteed loan. In columns 4 and 5 instead the sample is split between firms with a revolving credit line as of February 2020 and those whose credit did not include a revolving credit line. The estimates indicate that the interest rate reduction associated with 100% credit substitution is smaller for the former (60 bp) than for the latter (74 bp), consistent with the idea that in the former case firms face banks with greater bargaining power than in the latter, again in line with the model's predictions.

Finally, in Table 10, as done in Table 8 for interest rates, we present regressions whose dependent variable is the change in average residual maturity. The results are similar. For the average bank-firm relationship with a guaranteed credit, the increase in residual maturity associated with the guarantee is 9 months relative to other banks lending to the same firm. However, in this case, the presence of a bank-firm relationship appears to reduce (or even offset) the positive effect of the availability of the loan guarantee: the interaction with the share of granted credit has a negative and significant coefficient. Moreover, banks issuing guaranteed loans offer greater maturity extensions to firms that already have a longer residual maturity. The puzzle may be solved if relationship banks typically have longer maturity loans, so that in the estimates, residual maturity acts as the most relevant measure of the strength of the bank-firm relationship.

6.4. Firm-level estimates

Finally, we inquire whether the substitution observed at the bank-firm level also matters at the firm level, by investigating whether the size of the firm-level guarantee affected the overall amount of non-guaranteed credit received by firms included in loan guarantee programs. To this purpose, we estimate the specification of Eq. (14) at the firm level for the subsample of firms that received a guaranteed loan, including those borrowing from a single bank. While the resulting estimates of substitution are of interest as they consider the overall response of firm-level credit supply to loan guarantees, it is exposed to selection concerns because firm-level variation in the amount of loan guarantees is non-random, being dictated by eligibility rules.

The firm-level estimates are shown in Table 11 for the pooled sample of firms receiving guaranteed loans in all four countries, and in Table A8 separately in each country. The estimated coefficients of the L_i^G variable shown in Table 11 indicate that larger loan guarantees are associated with lower non-guaranteed firm-level credit, the magnitude of substitution ranging between 0.09 and 0.12 depending on the specification: on average a 1-euro increase in a firm's guaranteed lending is associated with substitution ranging between 9 and 12 cents, which translates into an average increase in total lending ranging between 88 and 91 cents. Even in this different specification, firm and bank heterogeneity associated with credit substitution have similar implications for credit substitution to those seen in previous tables. The coefficients of interactions between the guaranteed loan amount and firm-level characteristics show that the response of substitution to guarantees is larger for firms in more affected sectors and countries, and for smaller and riskier firms. While the coefficients are statistically significant for size and risk, they are not for value-added growth. In

Table 7
Substitution: firm-bank level analysis, including single-bank firms.

Dependent variable:	Credit substitution (s_{ij})				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	0.337***	0.342***	0.343***	0.326***	0.307***
	(0.028)	(0.025)	(0.019)	(0.021)	(0.019)
L_{ij}^G ×Industry VA Growth		0.141	0.114	0.103	0.081*
		(0.103)	(0.103)	(0.082)	(0.046)
L_{ij}^G ×Firm Size		−0.009	−0.007	−0.009	−0.026***
		(0.007)	(0.007)	(0.007)	(0.008)
L_{ij}^G ×Firm Risk		0.493***	0.463***	0.456***	0.453***
		(0.052)	(0.043)	(0.046)	(0.048)
L_{ij}^G ×Bank Assets		0.010	0.017	0.002	0.000
		(0.015)	(0.015)	(0.012)	(0.010)
L_{ij}^G ×Bank Liquidity		0.001***	0.001***	0.001**	0.001**
		(0.000)	(0.000)	(0.000)	(0.000)
L_{ij}^G ×Bank Capital		1.080		1.613**	1.258**
		(0.781)		(0.685)	(0.563)
L_{ij}^G ×Bank NPL			−2.450***	−3.138***	−2.169***
			(0.676)	(0.828)	(0.835)
L_{ij}^G ×Share of Granted					−0.322***
					(0.035)
L_{ij}^G ×Drawn/Granted					−0.079
					(0.117)
L_{ij}^G ×Residual Maturity					0.100***
					(0.033)
Industry-Location-Size FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R^2	0.130	0.126	0.127	0.130	0.157
N	3 561 108	2 555 452	2 563 497	2514 094	2102 075

This table reports bank-firm level estimates of an equation whose dependent variable is the credit substitution s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

this firm-level specification as well, credit substitution is significantly greater for firms borrowing from stronger banks, i.e., those featuring fewer NPLs, more liquidity and stronger capitalization, confirming the results in previous tables.

Table A8 of the Appendix presents the estimates for the most complete specification (shown in column 4 of Table 11) separately for each of the four countries. Larger loan guarantees are associated with lower non-guaranteed credit in all four countries, but the magnitude of the reduction varies across countries, being largest in Italy (0.16) and smallest in Spain (0.07): thus, on average a 1-euro increase in a firm's guaranteed lending is associated with substitution ranging between 7 and 16 cents depending on the country. Firm and bank characteristics appear to play a similar role across countries: in all of them, the extent of credit substitution is larger for weaker firms (i.e., smaller and riskier firms in more affected sectors and countries) borrowing from stronger banks (i.e., with fewer NPLs, larger, more capitalized and liquid banks). In some cases, the relevant coefficients are not statistically significant, but their signs are consistent across countries.

Table A9 reports estimates where the amount of each guaranteed loan is replaced by the corresponding guaranteed amount, calculated as the value of the loan multiplied by its guaranteed fraction. The results are similar to those shown in Table 11, although the average substitution is larger, reflecting the smaller size of the explanatory variable: the coefficient estimate rises from 0.11 (in column 1 of Table 11) to 0.20 (in column 1 of Table A9). The results about how substitution varies depending on firm and bank characteristics are also qualitatively unchanged.

7. Conclusions

This paper investigates whether public credit guarantee schemes, used extensively after the onset of the COVID-19 pandemic to support bank lending by shifting default risk to governments, led to substituting non-guaranteed with guaranteed credit. In principle, such substitution may be driven by banks exploiting public guarantees as an opportunity to reduce their pre-existing credit risk exposure, as well as by firms using them as an opportunity to reduce their credit costs. We provide a simple model of the effects of loan guarantees on banks' lending policies, showing that banks should engage more in credit substitution vis-à-vis riskier borrowers and should give them comparatively large interest rate concessions.

We take these predictions to the data by exploiting a novel harmonized credit register dataset for the entire euro area, AnaCredit, matched with supervisory bank balance-sheet data, and focus on the four largest euro area countries. We find that, on average, banks reduced non-guaranteed credit below its pre-pandemic level upon granting guaranteed lending. The point estimate of such credit substitution turns out to depend on various features of the methodology used, namely: (i) inclusion or exclusion from the sample of firms that received

Table 8
Firm-bank level analysis of changes in interest rates.

Dependent variable:	$\Delta InterestRate_{ij}$					
	(1)	(2)	(3)	(4)	(5)	(6)
L_{ij}^{G}	−0.387***	−0.449***	−0.488***	−0.515***	−0.559***	−0.426***
	(0.041)	(0.074)	(0.073)	(0.079)	(0.070)	(0.066)
L_{ij}^{G} ×Industry VA Growth		0.466***	0.397***	0.392***	0.452***	0.181***
		(0.127)	(0.121)	(0.120)	(0.118)	(0.060)
L_{ij}^{G} ×Firm Size		0.072***	0.076***	0.076***	0.066***	0.060***
		(0.016)	(0.017)	(0.016)	(0.016)	(0.016)
L_{ij}^{G} ×Firm Risk		0.149	0.118	0.103	−1.513**	−0.382**
		(0.442)	(0.431)	(0.449)	(0.674)	(0.181)
L_{ij}^{G} ×Bank Assets		0.043	0.027	0.026	0.035	0.027
		(0.028)	(0.026)	(0.029)	(0.029)	(0.021)
L_{ij}^{G} ×Bank Liquidity		0.034	0.033	0.038	0.034	0.024
		(0.127)	(0.126)	(0.129)	(0.133)	(0.132)
L_{ij}^{G} ×Bank Capital		0.119		0.838	0.463	1.746***
		(0.915)		(0.980)	(1.066)	(0.604)
L_{ij}^{G} ×Bank NPL			−3.145***	−3.172***	−4.399***	−2.871***
			(1.209)	(1.336)	(1.429)	(0.953)
L_{ij}^{G} ×Share of Granted					−1.170***	−0.773***
					(0.139)	(0.134)
L_{ij}^{G} ×Drawn/Granted					0.585***	0.457***
					(0.083)	(0.052)
L_{ij}^{G} ×Residual Maturity					−0.152*	−0.040
					(0.084)	(0.066)
L_{ij}^{G} ×Lending Rate						−37.480***
						(2.160)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes	Yes
R^2	0.388	0.402	0.402	0.403	0.412	0.728
N	767 159	557 527	557 383	547 697	474 338	469 215

This table reports bank-firm level estimates of an equation whose dependent variable is the $\Delta InterestRate_{ij}$ charged by bank j to firm i between February 2020 and August 2020. The main explanatory variable, L_{ij}^{G}, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^{G} and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1

Table 9
Reduction in interest rate and credit substitution.

Dependent variable:	$\Delta InterestRate_{ij}$				
	Full Sample	Firms borrowing from one bank	Firms borrowing from multiple banks	Firms with revolving credit	Firms without revolving credit
	(1)	(2)	(3)	(4)	(5)
s_{ij}	−0.788***	−0.281***	−1.090***	−0.596***	−0.738***
	(0.100)	(0.0718)	(0.146)	(0.106)	(0.138)
Bank FE	Yes	Yes	Yes	Yes	Yes
R^2	0.0539	0.0773	0.0493	0.0401	0.198
N	680 665	258 997	421 591	488 906	191 624

This table reports bank-firm level estimates of an equation whose dependent variable is the $\Delta InterestRate_{ij}$ charged by bank j to firm i between February 2020 and August 2020. The explanatory variable is credit substitution s_{ij}, defined as the negative of the change in non-guaranteed credit granted by bank j to firm i between February 2020 and August 2020, divided by total initial credit granted to firm i in February 2020. The sample is made by only those bank-firm relationships with a guaranteed loan between February 2020 and August 2020. We include both firms with single and multiple banks. In column 1 we consider the full sample. In columns 2 and 3 we split the sample between firms borrowing only from one bank (column 2) and firms borrowing from multiple banks (column 3). In columns 4 and 5 we split the sample between firms with at least one revolving credit instrument in February 2020 (column 4) and firms without revolving credit instruments in February 2020. Robust standard errors are in parenthesis. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 10
Firm-bank level analysis of changes in loan maturities.

Dependent variable:	$\Delta Maturity_{ij}$				
	(1)	(2)	(3)	(4)	(5)
L_{ij}^G	23.612***	25.754***	21.681***	21.513***	17.532***
	(2.031)	(1.556)	(2.176)	(1.931)	(2.325)
L_{ij}^G ×Industry VA Growth		−4.265	−2.081	−2.078	−3.358
		(5.058)	(5.555)	(5.406)	(3.937)
L_{ij}^G ×Firm Size		−0.255	−0.104	−0.0315	−0.139
		(0.414)	(0.429)	(0.436)	(0.560)
L_{ij}^G ×Firm Risk		16.742***	19.401***	19.374***	5.946
		(3.952)	(4.368)	(4.514)	(4.945)
L_{ij}^G ×Bank Assets		2.030***	0.300	0.459	0.817
		(0.598)	(0.796)	(0.721)	(0.819)
L_{ij}^G ×Bank Liquidity		0.018	0.037	0.037*	0.044**
		(0.017)	(0.024)	(0.021)	(0.020)
L_{ij}^G ×Bank Capital		−146.336***		−135.928***	−67.954***
		(25.961)		(28.433)	(19.752)
L_{ij}^G ×Bank NPL			347.172***	327.514***	201.025***
			(59.642)	(56.341)	(45.061)
L_{ij}^G ×Share of Granted					−54.382***
					(5.272)
L_{ij}^G ×Drawn/Granted					4.038*
					(2.202)
L_{ij}^G ×Residual Maturity					8.129***
					(1.115)
Firm FE	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	No	Yes	Yes	Yes	Yes
R^2	0.348	0.371	0.371	0.371	0.406
N	814 348	597 034	597 205	587 343	507 143

This table reports bank-firm level estimates of an equation whose dependent variable is the $\Delta Maturity_{ij}$ in the bank-firm relationship between bank j and firm i between February 2020 and August 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans; Share of granted, defined as the share of the bank j out of the total bank exposure of the firm i; Drawn/Granted, defined as the amount of credit drawn by firm i divided by the amount granted by bank j to firm i; Residual maturity, defined as the log of the number of remaining months until the expiration or the repayment of the credit by bank j to firm i. All the explanatory variables, apart from L_{ij}^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the bank and firm level, are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

the guaranteed loan from banks not previously lending to them: when these firms are included, we estimate a drop of 28 cents in non-guaranteed credit for 1 euro of guaranteed lending, against 46 cents when these firms are excluded; (ii) callability of preexisting credit: estimates range from 25 cents for term loans and non-revolving credit lines to 43 cents for revolving credit lines, which are callable; (iii) estimation at within-firm or at firm level: in the first case, where the estimate refers to the credit reduction by the bank issuing the guaranteed loan relative to other banks lending to the same firm, the estimate is 28 cents, while in the second, where it refers to the credit reduction by all banks lending to the recipient of the guaranteed loan, the estimate is 9 cents. Within this range, we consider the 28-cent within-firm estimate to be the most reliable: first, it is better identified than the firm-level estimate; second, including observations regarding loans extended by new banks appears appropriate, as loan guarantees enabled some firms to access novel lenders; third, this estimate aggregates that referring to non-callable and callable credit, appropriately accounting for the relatively modest weight of the latter.

Consistently with our model, we find credit substitution to be higher for guaranteed loans granted to riskier firms, i.e., those with more arrears, smaller sizes, and operating in more affected sectors and countries. Also in line with the model, the evidence reveals a positive correlation between the interest rate reduction offered to firms receiving guaranteed loans and credit substitution by the corresponding lenders: riskier and smaller firms operating in more affected sectors were given larger interest rate reductions upon receiving guaranteed loans. As a result, the availability of loan guarantees generates a tradeoff: in bank-firm relationships featuring more credit substitution, firms obtain comparatively more generous interest rate concessions. Consistent with the model, we find that this tradeoff is affected by the distribution of bargaining power between banks and firms: when the bank providing the guaranteed loan is the sole firm's lender as opposed to one of several lenders, and when formerly lent to the firm only in the form of a callable credit line as opposed to non-callable credit, it grants a much smaller interest rate concession to the firm for given amount of credit substitution.

Instead, banks with stronger balance sheets, i.e., with more capital and/or fewer NPLs, featured both greater substitution and smaller interest rate reductions upon offering guaranteed loans. The finding that these banks transferred less of the benefits from loan guarantees to client firms via reduced cost of credit is consistent with the model's predictions if these banks feature greater bargaining power in renegotiating their initial loans.

Overall, the evidence suggests that, in the euro area, loan guarantees contributed to the continued extension of credit to relatively creditworthy firms, but also benefited the balance sheet of banks, especially the comparatively stronger ones. It also highlights that the provision of credit to weaker firms was more severely hampered by credit substitution, although these firms obtained comparatively large interest rate cuts and loan maturity extensions.

Table 11
Credit substitution: firm-level analysis.

Dependent variable:	Credit substitution (s_i)				
	(1)	(2)	(3)	(4)	5
L_i^G	0.107***	0.120***	0.113***	0.095***	0.086***
	(0.021)	(0.026)	(0.020)	(0.013)	(0.013)
L_i^G x Industry VA Growth		−0.041	−0.052	−0.013	−0.018
		(0.060)	(0.052)	(0.055)	(0.037)
L_i^G x Firm Size		−0.023***	−0.025***	−0.020***	−0.022***
		(0.006)	(0.007)	(0.006)	(0.006)
L_i^G x Firm Risk		0.340***	0.352***	0.328***	0.318***
		(0.057)	(0.061)	(0.052)	(0.057)
L_i^G x Bank assets			0.001	0.006	0.010
			(0.012)	(0.012)	(0.010)
L_i^G x Bank Liquidity			0.001***	0.000**	0.002***
			(0.000)	(0.000)	(0.000)
L_i^G x Bank Capital			1.222*		1.639***
			(0.625)		(0.511)
L_i^G x Bank NPL				−2.870***	−3.352***
				(0.487)	(0.550)
Country FE	Yes	Yes	Yes	Yes	Yes
Non-interacted variables	Yes	Yes	Yes	Yes	Yes
R^2	0.034	0.040	0.053	0.046	0.060
N	623 133	500 013	457 697	460 275	456 306

The table reports firm-level estimates of a regression whose dependent variable is the credit substitution s_i, defined as the negative of the change in non-guaranteed credit received by firm i between February 2020 and August 2020, divided by its total credit as of February 2020. The main explanatory variable, L_{ij}^G, is the amount of government-guaranteed credit received by the firm, divided by total credit granted to the firm as of February 2020. Other explanatory variables are: Industry VA Growth, defined as the industrial sector change in Valued Added in each country between February 2020 and August 2020; Firm Size, proxied by the log of firm total debt; Firm Risk, proxied by the share of loans in arrears out of total loans; Bank Assets, defined as the log of total bank assets; Bank Liquidity, defined as the bank Liquidity Coverage Ratio; Bank Capital, defined as the Core Tier 1 Ratio and Bank NPL, defined as the share of NPL loans out of its total loans. Each of the bank related variables is calculated as a weighted average of the corresponding bank-level variable, where the weights are the shares of the banks' exposure toward the firm out of total bank exposure of the firm at December 2019. All the explanatory variables, apart from L_i^G and Industry VA Growth, are measured as of December 2019. Standard errors, double-clustered at the main bank level, are in parentheses. *** p < 0.01, ** p < 0.05, * p<0.1.

CRediT authorship contribution statement

Carlo Altavilla: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrew Ellul:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Marco Pagano:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Andrea Polo:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Thomas Vlassopoulos:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2025.104137.

Data availability

The original data is available at: https://data.mendeley.com/datasets/34pj2x9r7z/2.

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